

INTERNATIONAL COLLEGIATE PROGRAMMING CONTEST
ACM-ICPC REFERENCE MANUAL

Team Notebook

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Table of Contents

1	Environment Setup	2
1.1	Checklist & Code::Blocks Setup	2
1.2	Template & Fast I/O	3
1.3	Optimization & Diagnostic Pragmas	3
1.4	Custom Fast I/O Extensions	3
2	Data Structures	3
2.1	Chtholly Tree (Old Driver Tree)	3
2.2	DSU with Rollback	3
2.3	Binary Grouping for Dynamic Insertion	3
2.4	Fenwick Tree with lower/upper bound	3
2.5	Mo's Algorithm	4
2.6	Mo's Algorithm on Trees	4
2.7	Mo's Algorithm with Updates (3D Mo)	4
2.8	Segment Tree 2N (Iterative)	5
2.9	Persistent Segment Tree	5
2.10	Treap	5
2.11	Implicit Treap	5
2.12	Wavelet Tree	6
3	Graphs and Trees	6
3.1	2-SAT Solver	6
3.2	Articulation Points & Bridges	6
3.3	Centroid Decomposition	6
3.4	Dijkstra Algorithm (Dense Graph)	6
3.5	Finding Bridges Online	7
3.6	Dinic Algorithm	7
3.7	Gomory-Hu Tree	7
3.8	Min Cost Max Flow (MCMF)	8
3.9	Flow with Lower Bound	8
3.10	Heavy-Light Decomposition (HLD)	8
3.11	Hopcroft-Karp Algorithm	8
3.12	Hungarian Algorithm (Kuhn-Munkres)	9
3.13	Kruskal Reconstruction Tree (KRT)	9
3.14	Link-Cut Tree	9
3.15	Boruvka's Algorithm	10
3.16	Prim's Algorithm	10
3.17	Tarjan Algorithm	10
3.18	Virtual Tree (Auxiliary Tree)	10
4	Math	11
4.1	Chinese Remainder Theorem (CRT)	11
4.2	Euler's Totient Function (Phi)	11
4.3	Extended Euclidean	11

4.4	FFT & NTT	11
4.5	Matrix Exponentiation Template	12
4.6	Miller-Rabin Primality Test & Pollard's Rho	12
4.7	Binary GCD (Stein's Algorithm)	12
4.8	Stern-Brocot Tree	12
4.9	Tonelli-Shanks Algorithm	12
5	Strings	13
5.1	Aho-Corasick Automaton	13
5.2	KMP Algorithm	13
5.3	Manacher's Algorithm	13
5.4	Palindrome Tree (Eertree)	13
5.5	String Double Hashing	14
5.6	Suffix Array	14
5.7	Suffix Automaton (SAM)	14
5.8	Trie	14
5.9	Z-Algorithm	14
6	Bit Manipulation	15
6.1	Binary Trie (Bit Trie)	15
6.2	Bitwise Hacks	15
6.3	XOR Basis	15
7	DPs	15
7.1	Aliens Trick (WQS Binary Search)	15
7.2	Bitmask DP for Set Partitioning	15
7.3	Divide & Conquer Optimization	15
7.4	Knuth Optimization	16
7.5	Sum Over Subsets DP (SOS DP)	16
8	Geometry	16
8.1	Geometry Templates (Integer & Double)	16
8.2	Convex Hull (Andrew's Monotone Chain)	16
8.3	Incenter & Circumcenter of a Triangle	16
8.4	Li Chao Tree	16
8.5	LineContainer (CHT)	17
8.6	Polygon Area (Shoelace Formula)	17
8.7	Smallest Enclosing Circle (Welzl's Algorithm)	17
9	Misc	17
9.1	Custom Hash for hashmap & hashset	17
9.2	Hill Climbing (Heuristics)	17
9.3	Simulated Annealing (Heuristics)	17
9.4	ModInt Template	18
9.5	Policy-Based Data Structures (PBDS)	18
9.6	Stress Tester	18
10	Classic	18
10.1	Bounded Knapsack with Binary Grouping	18

10.2	CSES Counting Tilings	19
10.3	Closest Pair of Points	19
10.4	De Bruijn Sequence	19
11	Formulas	19
11.1	Essential Theorems	19
11.2	Essential Formulas	19
11.3	Cheatsheet	20

1 Environment Setup

1.1 Checklist & Code::Blocks Setup

Contest preparation, environment validation, and shortcuts.
Contest Preparation & Environment Verification:

- Bring ID, credentials, pens, and printed Team Notebook.
- Verify system time, keyboard layout, and workspace comfort.
- Enable C++17/14:
Settings -> Compiler -> Compiler Settings -> Flag "-std=c++17".
- Add compiler optimization & warning flags: -O3, -Wall, -Wextra, -Wshadow.
- Enable auto-save:
Settings -> Environment -> Autosave -> Check every 1 min.
- Snippets: Settings -> Editor -> Abbreviations. Expands with Ctrl + J.

Essential Code::Blocks Shortcuts:

- Ctrl + Shift + C: Comment selected block.
- Ctrl + Shift + X: Uncomment selected block.
- Ctrl + F9 / Ctrl + F10 / F9: Compile / Run / Both.
- Ctrl + Space / Ctrl + Shift + Space: Autocomplete / Parameter tips.
- Alt + G: Jump to declaration or definition.
- Alt + Up/Down: Move selected lines up or down.
- Ctrl + E: Select next occurrence of keyword (Multi-select).
- Ctrl + Click: Multi-edit at custom positions.
- Ctrl + D: Duplicate selected text or current line.
- Ctrl + T: Swap current line with the line above it.

1.2 Template & Fast I/O

Standard execution template with optimized input/output streams for competitive programming.

```
#include <bits/stdc++.h>
using namespace std;

void solve() {

}

int main() {
    ios_base::sync_with_stdio(0); cin.tie(0);
    int t = 1; cin >> t;
    while (t--) solve();
    return 0;
}
```

1.3 Optimization & Diagnostic Pragmas

Enables SIMD vectorization, loop unrolling, and silences diagnostic warnings on older judge systems. Place at the absolute top of the source file.

```
#pragma GCC optimize("O3,unroll-loops")
#pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")
#pragma GCC diagnostic ignored "-Wpragmas"
#pragma GCC diagnostic ignored "-Wunused-result"
#pragma GCC diagnostic ignored "-Wunknown-pragmas"
```

1.4 Custom Fast I/O Extensions

Low-level fast I/O using unlocked character reading. Handles positive integers. Returns -1 on EOF.

```
#ifdef _WIN32
#define GETCHAR _getchar_nolock
#else
#define GETCHAR getchar_unlocked
#endif

template<typename T = int>
inline T readNum() {
    T x = 0; char ch = GETCHAR();
    while (ch < '0' || ch > '9') { if (ch == EOF) return -1; ch = GETCHAR(); }
    while (ch >= '0' && ch <= '9') { x = (x << 3) + (x << 1) + (ch - '0'); ch = GETCHAR(); }
    return x;
}
```

2 Data Structures

2.1 Chtholly Tree (Old Driver Tree)

```
struct Node {
    int l, r;
    mutable long long val;

    Node(int l, int r = -1, long long val = 0) : l(l), r(r), val(val) {}

    bool operator<(const Node& o) const { return l < o.l; }
};
```

```
struct ChthollyTree {
    set<Node> odt;

    void init(const vector<long long>& a) {
        odt.clear();
        for (int i = 1; i < a.size(); i++) {
            odt.insert(Node(i, i, a[i]));
        }
    }

    auto split(int pos) {
        auto it = odt.lower_bound(Node(pos));
        if (it != odt.end() && it->l == pos) return it;

        --it;
        if (it->r < pos) return odt.end();

        int l = it->l, r = it->r;
        long long v = it->val;

        odt.erase(it);
        odt.insert(Node(l, pos - 1, v));
        return odt.insert(Node(pos, r, v)).first;
    }

    void assign(int l, int r, long long val) {
        auto itr = split(r + 1); // split(r + 1) first
        auto itl = split(l);

        odt.erase(itl, itr);
        odt.insert(Node(l, r, val));
    }

    void add_range(int l, int r, long long val) {
        auto itr = split(r + 1);
        auto itl = split(l);

        for (; itl != itr; ++itl) {
            itl->val += val;
        }
    }
};
```

2.2 DSU with Rollback

Complexity: $O(\log N)$. Path compression disabled. Use `get_time()` to save checkpoint and `rollback(t)` to revert.

```
struct DSU_Rollback {
    vector<int> parent, sz;
    vector<pair<int, int>> history;

    DSU_Rollback(int n) {
        parent.resize(n + 1); sz.assign(n + 1, 1);
        iota(parent.begin(), parent.end(), 0);
    }

    int find_set(int v) {
        while (v != parent[v]) v = parent[v];
        return v;
    }

    int get_time() { return history.size(); }

    bool union_sets(int a, int b) {
        a = find_set(a); b = find_set(b);
        if (a == b) return false;

        if (sz[a] < sz[b]) swap(a, b);
        history.push_back({b, parent[b]});
        history.push_back({a, sz[a]});
```

```
        parent[b] = a; sz[a] += sz[b];
        return true;
    }

    void rollback(int t) {
        while (history.size() > t) {
            auto [u, val] = history.back(); history.pop_back();
            if (u < 0) sz[~u] = val;
            else parent[u] = val;
        }
    }
};
```

2.3 Binary Grouping for Dynamic Insertion

Converts static data structures (build $O(f(N))$) into dynamic online variants supporting insertions in amortized $O(\frac{f(N) \log N}{N})$ and queries in $O(g(N) \log N)$.

```
struct StaticACAM {
    vector<string> strings;
    void insert(const string& s) { strings.push_back(s); }
    int size() const { return strings.size(); }
    void build() {}
    long long query(const string& s) { return 0; }
};

struct DynamicACAM {
    vector<StaticACAM> acams;

    void insert(const string& s) {
        StaticACAM cur; cur.insert(s);
        acams.push_back(move(cur));
        while (acams.size() >= 2 && acams.back().size() == acams[acams.size() - 2].size()) {
            StaticACAM rhs = move(acams.back()); acams.pop_back();
            StaticACAM lhs = move(acams.back()); acams.pop_back();
            for (const auto& str : rhs.strings) lhs.insert(str);
            lhs.build();
            acams.push_back(move(lhs));
        }
    }

    long long query(const string& s) {
        long long ans = 0;
        for (auto& ac : acams) ans += ac.query(s);
        return ans;
    }
};
```

Handles dynamic element deletions by initializing two separate DynamicACAM instances (one for insertions, one for deletions) and tracking the net difference.

2.4 Fenwick Tree with lower/upper bound

$O(\log N)$ binary lifting to find the first index where prefix sum $\geq$ or $>$ target. Works only for non-negative frequencies.

```
struct BIT {
    int n; vector<int> tree;
    BIT(int n) : n(n), tree(n + 1, 0) {}

    void update(int i, int delta) {
        for (; i <= n; i += i & -i) tree[i] += delta;
    }

    int lower_bound(int sum) {
```

```

int idx = 0;
for (int i = 1 << __lg(n); i > 0; i >= 1) {
    if (idx + i <= n && tree[idx + i] < sum) {
        idx += i; sum -= tree[idx];
    }
}
return idx + 1;
}

int upper_bound(int sum) {
int idx = 0;
for (int i = 1 << __lg(n); i > 0; i >= 1) {
    if (idx + i <= n && tree[idx + i] <= sum) {
        idx += i; sum -= tree[idx];
    }
}
return idx + 1;
}
};

```

2.5 Mo's Algorithm

Offline queries sorted by $\sqrt{N}$ -sized blocks. Total time $O((N + Q)\sqrt{N})$. Set $BLOCK_SIZE \approx N/\sqrt{Q}$.

```

const int BLOCK_SIZE = 450;

struct Query {
    int l, r, id;
    bool operator<(const Query& other) const {
        int b1 = l / BLOCK_SIZE, b2 = other.l / BLOCK_SIZE;
        if (b1 != b2) return b1 < b2;
        return (b1 & 1) ? (r < other.r) : (r > other.r);
    }
};

long long current_answer = 0;
vector<int> cnt;

void add(int idx, const vector<int>& a) {}
void remove(int idx, const vector<int>& a) {}

vector<long long> solve_mo(vector<Query>& queries, const vector<int>& a) {
    sort(queries.begin(), queries.end());
    cnt.assign(1000005, 0); current_answer = 0;

    vector<long long> answers(queries.size());
    int cur_l = 0, cur_r = -1;

    for (const auto& q : queries) {
        while (cur_l > q.l) add(--cur_l, a);
        while (cur_r < q.r) add(++cur_r, a);
        while (cur_l < q.l) remove(cur_l++, a);
        while (cur_r > q.r) remove(cur_r--, a);
        answers[q.id] = current_answer;
    }
    return answers;
}

```

2.6 Mo's Algorithm on Trees

Offline queries on trees using Euler Tour (flattening the tree into an array). Set $BLOCK_SIZE \approx \frac{2N}{\sqrt{Q}}$ for an optimal time complexity of $O((N + Q)\sqrt{N})$.

```

const int BLOCK = 450;
const int MAXN = 100005;

```

```

struct Query {
    int l, r, lc, id;
    bool operator<(const Query& o) const {
        int b1 = l / BLOCK, b2 = o.l / BLOCK;
        if (b1 != b2) return b1 < b2;
        return (b1 & 1) ? (r < o.r) : (r > o.r);
    }
};

int n, q, timer;
int st[MAXN], en[MAXN], id[2 * MAXN], depth[MAXN], up[MAXN][20];
vector<int> adj[MAXN];
bool vis[MAXN];

void dfs(int u, int p) {
    st[u] = ++timer; id[timer] = u;
    up[u][0] = p;
    for (int i = 1; i < 20; ++i) up[u][i] = up[up[u][i-1]][i-1];
    for (int v : adj[u]) {
        if (v != p) { depth[v] = depth[u] + 1; dfs(v, u); }
    }
    en[u] = ++timer; id[timer] = u;
}

int get_lca(int u, int v) {
    if (depth[u] < depth[v]) swap(u, v);
    for (int i = 19; i >= 0; --i) {
        if (depth[u] - (1 << i) >= depth[v]) u = up[u][i];
    }
    if (u == v) return u;
    for (int i = 19; i >= 0; --i) {
        if (up[u][i] != up[v][i]) { u = up[u][i]; v = up[v][i]; }
    }
    return up[u][0];
}

void check(int u) {
    if (vis[u]) { /* remove_from_data_structure(u); */ }
    else { /* add_to_data_structure(u); */ }
    vis[u] ^= 1;
}

vector<long long> solve_mo_tree(vector<Query>& queries) {
    timer = 0; dfs(1, 1);
    for (auto& q : queries) {
        int u = q.l, v = q.r; // Here q.l and q.r store original
        // tree node indices
        int lca = get_lca(u, v);
        if (st[u] > st[v]) swap(u, v);
        if (lca == u) {
            q.l = st[u]; q.r = st[v]; q.lc = 0;
        } else {
            q.l = en[u]; q.r = st[v]; q.lc = lca;
        }
    }
    sort(queries.begin(), queries.end());

    vector<long long> answers(queries.size());
    int cur_l = 1, cur_r = 0;
    for (const auto& q : queries) {
        while (cur_l > q.l) check(id[--cur_l]);
        while (cur_r < q.r) check(id[++cur_r]);
        while (cur_l < q.l) check(id[cur_l++]);
        while (cur_r > q.r) check(id[cur_r--]);
        if (q.lc) check(q.lc);
        answers[q.id] = 0; // fetch current answer here
        if (q.lc) check(q.lc);
    }
    return answers;
}

```

- Convert path queries into range queries on the `id` array size $2N$ (1-indexed).

- If LCA is one of the endpoints, the range is `[st[u], st[v]]`. Otherwise, the range is `[en[u], st[v]]` plus the isolated `lca` node.
- The function `check(u)` toggles the presence of node u , handling insertions and deletions symmetrically.

2.7 Mo's Algorithm with Updates (3D Mo)

Offline queries with point updates sorted by block-based 3D coordinates. Set $BLOCK_SIZE \approx N^{2/3}$ for an optimal time complexity of $O(N^{5/3})$.

```

const int BLOCK = 2150;

struct Query {
    int l, r, t, id;
    bool operator<(const Query& o) const {
        int b11 = l / BLOCK, b12 = o.l / BLOCK;
        if (b11 != b12) return b11 < b12;
        int b21 = r / BLOCK, b22 = o.r / BLOCK;
        if (b21 != b22) return b21 < b22;
        return t < o.t;
    }
};

struct Update {
    int pos, old_val, new_val;
};

long long current_answer = 0;
vector<int> cnt;

void add(int idx, const vector<int>& a) {}
void remove(int idx, const vector<int>& a) {}

void apply_update(int t, int cur_l, int cur_r, vector<int>& a, const vector<Update>& updates) {
    int pos = updates[t].pos;
    if (pos >= cur_l && pos <= cur_r) {
        remove(pos, a);
        a[pos] = updates[t].new_val;
        add(pos, a);
    } else {
        a[pos] = updates[t].new_val;
    }
}

void rollback_update(int t, int cur_l, int cur_r, vector<int>& a, const vector<Update>& updates) {
    int pos = updates[t].pos;
    if (pos >= cur_l && pos <= cur_r) {
        remove(pos, a);
        a[pos] = updates[t].old_val;
        add(pos, a);
    } else {
        a[pos] = updates[t].old_val;
    }
}

vector<long long> solve_mo_update(vector<Query>& queries, vector<Update>& updates, vector<int>& a) {
    sort(queries.begin(), queries.end());
    cnt.assign(1000005, 0); current_answer = 0;
    vector<long long> answers(queries.size());

    int cur_l = 0, cur_r = -1, cur_t = 0;
    for (const auto& q : queries) {
        while (cur_t < q.t) apply_update(cur_t++, cur_l, cur_r, a, updates);
    }
}

```

```

    while (cur_t > q.t) rollback_update(--cur_t, cur_l, cur_r,
a, updates);
    while (cur_l > q.l) add(--cur_l, a);
    while (cur_r < q.r) add(++cur_r, a);
    while (cur_l < q.l) remove(cur_l++, a);
    while (cur_r > q.r) remove(cur_r--, a);
    answers[q.id] = current_answer;
}
return answers;
}

```

- Separate timeline modifications into the `updates` vector (0-indexed). Assign the total number of updates prior to a query to `Query.t`.

- Tune `BLOCK` dynamically based on N and Q to fit within the time limit.

2.8 Segment Tree 2N (Iterative)

0-indexed, $O(\log N)$ point update and range query $[l, r]$. Order matters for non-commutative operations. For 1-indexed, use size $2 * (n + 1)$.

```

struct SegTree {
    int n; vector<int> tree;
    SegTree(int n) : n(n), tree(2 * n, 0) {}

    void update(int i, int value) {
        for (tree[i += n] = value; i > 1; i >>= 1) {
            tree[i >> 1] = tree[i] + tree[i ^ 1];
        }
    }

    int query(int l, int r) {
        int res = 0;
        for (l += n, r += n + 1; l < r; l >>= 1, r >>= 1) {
            if (l & 1) res += tree[l++];
            if (r & 1) res += tree[--r];
        }
        return res;
    }
};

```

2.9 Persistent Segment Tree

$O(\log N)$ time and space per update. `upgrade(v, pos, val)` creates a new version from version v . Query version v via

`query(roots[v], 1, n, ql, qr)`.

```

struct Node {
    int val; Node *l, *r;
    Node(int val = 0) : val(val), l(nullptr), r(nullptr) {}
    Node(Node* o) : val(o->val), l(o->l), r(o->r) {}
};

struct PersistentSegTree {
    int n; vector<Node*> roots;
    PersistentSegTree(int n) : n(n) { roots.push_back(build(1, n)); }

    Node* build(int l, int r) {
        Node* curr = new Node(0);
        if (l == r) return curr;
        int mid = (l + r) >> 1;
        curr->l = build(l, mid); curr->r = build(mid + 1, r);
    }
};

```

```

    return curr;
}

Node* update(Node* prev_node, int l, int r, int pos, int val) {
    Node* curr = new Node(prev_node);
    if (l == r) { curr->val += val; return curr; }
    int mid = (l + r) >> 1;
    if (pos <= mid) curr->l = update(prev_node->l, l, mid, pos, val);
    else curr->r = update(prev_node->r, mid + 1, r, pos, val);
    curr->val = curr->l->val + curr->r->val;
    return curr;
}

int query(Node* node, int l, int r, int ql, int qr) {
    if (ql <= l && r <= qr) return node->val;
    int mid = (l + r) >> 1, res = 0;
    if (ql <= mid) res += query(node->l, l, mid, ql, qr);
    if (qr > mid) res += query(node->r, mid + 1, r, ql, qr);
    return res;
}

void upgrade(int version, int pos, int val) {
    roots.push_back(update(roots[version], 1, n, pos, val));
}
};

```

2.10 Treap

Balanced BST combining BST and Heap properties.

Operations run in $O(\log N)$ expected time.

```

struct Node {
    int key, prior, sz;
    Node *l, *r;
    Node(int key) : key(key), prior(rng()), sz(1), l(nullptr), r(
        nullptr) {}
};

int get_sz(Node* t) { return t ? t->sz : 0; }
void upd_sz(Node* t) { if (t) t->sz = 1 + get_sz(t->l) + get_sz(t->r); }

void split(Node* t, int key, Node*& l, Node*& r) {
    if (!t) l = r = nullptr;
    else if (t->key <= key) {
        split(t->r, key, t->r, r);
        l = t; upd_sz(l);
    } else {
        split(t->l, key, l, t->l);
        r = t; upd_sz(r);
    }
}

void merge(Node*& t, Node* l, Node* r) {
    if (!l || !r) t = l ? l : r;
    else if (l->prior > r->prior) {
        merge(l->r, l->r, r);
        t = l; upd_sz(t);
    } else {
        merge(r->l, l, r->l);
        t = r; upd_sz(t);
    }
}

void insert(Node*& t, Node* it) {
    if (!t) t = it;
    else if (it->prior > t->prior) {
        split(t, it->key, t->l, t->r);
        t = it; upd_sz(t);
    } else {
        insert(t->key <= it->key ? t->r : t->l, it);
    }
}

```

```

    upd_sz(t);
}

void erase(Node*& t, int key) {
    if (!t) return;
    if (t->key == key) {
        Node* th = t; merge(t, t->l, t->r); delete th;
    } else {
        erase(key < t->key ? t->l : t->r, key); upd_sz(t);
    }
}

```

2.11 Implicit Treap

Indexed by array positions instead of keys. Supports range operations (e.g., reverse) and split/merge in $O(\log N)$ expected time.

```

struct Node {
    int val, prior, sz;
    bool rev;
    Node *l, *r;
    Node(int val) : val(val), prior(rng()), sz(1), rev(false), l(
        nullptr), r(nullptr) {}
};

int get_sz(Node* t) { return t ? t->sz : 0; }
void upd_sz(Node* t) { if (t) t->sz = 1 + get_sz(t->l) + get_sz(t->r); }

void push(Node* t) {
    if (t && t->rev) {
        t->rev = false;
        swap(t->l, t->r);
        if (t->l) t->l->rev ^= true;
        if (t->r) t->r->rev ^= true;
    }
}

void split(Node* t, int k, Node*& l, Node*& r) {
    if (!t) { l = r = nullptr; return; }
    push(t);
    if (get_sz(t->l) < k) {
        split(t->r, k - get_sz(t->l) - 1, t->r, r);
        l = t; upd_sz(l);
    } else {
        split(t->l, k, l, t->l);
        r = t; upd_sz(r);
    }
}

void merge(Node*& t, Node* l, Node* r) {
    push(l); push(r);
    if (!l || !r) t = l ? l : r;
    else if (l->prior > r->prior) {
        merge(l->r, l->r, r);
        t = l; upd_sz(t);
    } else {
        merge(r->l, l, r->l);
        t = r; upd_sz(t);
    }
}

void reverse_range(Node*& t, int ql, int qr) {
    Node *l1, *r1, *l2, *r2;
    split(t, ql, l1, r1);
    split(r1, qr - ql + 1, l2, r2);
    if (l2) l2->rev ^= true;
    merge(r1, l2, r2);
    merge(t, l1, r1);
}

```

2.12 Wavelet Tree

Generalized range queries (k -th element, count $\leq k$) in $O(\log(\max A_i))$. Queries expect 1-indexed boundaries.

```
struct WaveletTree {
    int lo, hi;
    WaveletTree *l, *r;
    vector<int> b;

    template<typename Iter>
    WaveletTree(Iter from, Iter to, int lo, int hi) : lo(lo), hi(hi)
    , l(nullptr), r(nullptr) {
        if (from >= to || lo == hi) return;
        int mid = (lo + hi) >> 1;
        auto f = [mid](int x) { return x <= mid; };
        b.reserve(to - from + 1); b.push_back(0);
        for (auto it = from; it != to; ++it) b.push_back(b.back() + f(*it));
        auto pivot = stable_partition(from, to, f);
        l = new WaveletTree(from, pivot, lo, mid);
        r = new WaveletTree(pivot, to, mid + 1, hi);
    }

    int kth(int l, int r, int k) {
        if (lo == hi) return lo;
        int in_left = b[r] - b[l - 1], lb = b[l - 1], rb = b[r];
        if (k <= in_left) return l->kth(lb + 1, rb, k);
        return r->kth(l - lb, r - rb, k - in_left);
    }

    int count_le(int l, int r, int k) {
        if (l > r || k < lo) return 0;
        if (hi <= k) return r - l + 1;
        int lb = b[l - 1], rb = b[r];
        return l->count_le(lb + 1, rb, k) + r->count_le(l - lb, r - rb, k);
    }

    ~WaveletTree() { delete l; delete r; }
};
```

3 Graphs and Trees

3.1 2-SAT Solver

Solves 2-SAT in $O(V + E)$ using Tarjan's SCC. Variables are 1-indexed. Use negative sign for negation (e.g., -u).

```
struct TwoSat {
    int n, timer, scc_count;
    vector<vector<int>> adj;
    vector<int> tin, low, scc_id;
    vector<bool> in_stack, answer;
    stack<int> st;

    TwoSat(int n) : n(n), adj(2 * n + 1), tin(2 * n + 1, -1), low(2 * n + 1, -1),
        scc_id(2 * n + 1, -1), in_stack(2 * n + 1, false), answer(n + 1, false),
        timer(0), scc_count(0) {}

    void add_clause(int u, int v) {
        int not_u = (u > 0) ? u + n : -u;
        int act_u = (u > 0) ? u : -u + n;
        int not_v = (v > 0) ? v + n : -v;
        int act_v = (v > 0) ? v : -v + n;
        adj[not_u].push_back(act_v); adj[not_v].push_back(act_u);
    }
};
```

```
void dfs(int u) {
    tin[u] = low[u] = ++timer;
    st.push(u); in_stack[u] = true;
    for (int v : adj[u]) {
        if (tin[v] == -1) {
            dfs(v); low[u] = min(low[u], low[v]);
        } else if (in_stack[v]) {
            low[u] = min(low[u], tin[v]);
        }
    }
    if (low[u] == tin[u]) {
        scc_count++;
        while (true) {
            int v = st.top(); st.pop(); in_stack[v] = false;
            scc_id[v] = scc_count;
            if (u == v) break;
        }
    }
}

bool solve() {
    for (int i = 1; i <= 2 * n; i++) if (tin[i] == -1) dfs(i);
    for (int i = 1; i <= n; i++) {
        if (scc_id[i] == scc_id[i + n]) return false;
        answer[i] = scc_id[i] < scc_id[i + n];
    }
    return true;
}

};
```

3.2 Articulation Points & Bridges

Finds cut vertices and bridges in $O(V + E)$ using DFS. Handles multiple edges correctly.

```
struct CutVerticesBridges {
    int n, timer;
    vector<vector<int>> adj;
    vector<int> tin, low;
    vector<bool> is_cut;
    vector<pair<int, int>> bridges;

    CutVerticesBridges(int n, const vector<vector<int>>& adj) :
        n(n), adj(adj), timer(0), tin(n + 1, -1), low(n + 1, -1),
        is_cut(n + 1, false) {}

    void dfs(int u, int p = -1) {
        tin[u] = low[u] = ++timer;
        int children = 0, p_count = 0;

        for (int v : adj[u]) {
            if (v == p && ++p_count == 1) continue; // Handles multiple edges
            if (tin[v] != -1) {
                low[u] = min(low[u], tin[v]);
            } else {
                dfs(v, u);
                low[u] = min(low[u], low[v]);
                if (low[v] >= tin[u] && p != -1) is_cut[u] = true;
                if (low[v] > tin[u]) bridges.push_back({min(u, v), max(u, v)});
                children++;
            }
        }
        if (p == -1 && children > 1) is_cut[u] = true;
    }

    void run() {
        for (int i = 1; i <= n; i++) if (tin[i] == -1) dfs(i);
    }
};
```

```
};
```

3.3 Centroid Decomposition

Builds a centroid tree in $O(N \log N)$ time with depth $O(\log N)$. Tree layout is stored in parent for upward updates.

```
struct CentroidDecomposition {
    int n; vector<vector<int>> adj;
    vector<int> sz, parent; vector<bool> removed;

    CentroidDecomposition(int n, const vector<vector<int>>& tree_adj) :
        n(n), adj(tree_adj), sz(n + 1), parent(n + 1, 0), removed(n + 1, false) {}

    int get_sz(int u, int p) {
        sz[u] = 1;
        for (int v : adj[u]) if (v != p && !removed[v]) sz[u] += get_sz(v, u);
        return sz[u];
    }

    int find_centroid(int u, int p, int comp_sz) {
        for (int v : adj[u]) {
            if (v != p && !removed[v] && sz[v] > comp_sz / 2)
                return find_centroid(v, u, comp_sz);
        }
        return u;
    }

    int build_tree(int u) {
        int comp_sz = get_sz(u, 0), centroid = find_centroid(u, 0, comp_sz);
        removed[centroid] = true;
        for (int v : adj[centroid]) {
            if (!removed[v]) parent[build_tree(v)] = centroid;
        }
        return centroid;
    }

    void init(int start_node = 1) { build_tree(start_node); }
};
```

3.4 Dijkstra Algorithm (Dense Graph)

Shortest path algorithm for dense graphs ($E \approx V^2$) running in $O(V^2)$ without using a priority queue.

```
const long long INF = 1e18;

vector<long long> dijkstra_dense(int source, int n, const vector<vector<long long>>& adj) {
    vector<long long> dist(n + 1, INF);
    vector<bool> visited(n + 1, false);

    dist[source] = 0;
    for (int i = 1; i <= n; i++) {
        int u = -1;
        for (int v = 1; v <= n; v++) {
            if (!visited[v] && (u == -1 || dist[v] < dist[u])) u = v;
        }

        if (dist[u] == INF) break;
        visited[u] = true;

        for (int v = 1; v <= n; v++) {
            if (adj[u][v] != INF && dist[u] + adj[u][v] < dist[v]) {
                dist[v] = dist[u] + adj[u][v];
            }
        }
    }
}
```

```

        dist[v] = dist[u] + adj[u][v];
    }
}
return dist;
}

```

3.5 Finding Bridges Online

Maintains 2-edge-connected components dynamically under edge insertions in $O(N \log N + M \log N)$ total time using two DSU structures.

```

struct DynamicBridge {
    int bridges, lca_iter;
    vector<int> par, dsu_2ecc, dsu_cc, dsu_cc_size, last_visit;

    DynamicBridge(int n) : bridges(0), lca_iter(0), par(n),
        dsu_2ecc(n), dsu_cc(n), dsu_cc_size(n, 1), last_visit(n, 0) {
        iota(dsu_2ecc.begin(), dsu_2ecc.end(), 0);
        iota(dsu_cc.begin(), dsu_cc.end(), 0);
        fill(par.begin(), par.end(), -1);
    }

    int find_2ecc(int v) {
        if (v == -1) return -1;
        return dsu_2ecc[v] == v ? v : dsu_2ecc[v] = find_2ecc(
            dsu_2ecc[v]);
    }

    int find_cc(int v) {
        v = find_2ecc(v);
        return dsu_cc[v] == v ? v : dsu_cc[v] = find_cc(dsu_cc[v]);
    }

    void make_root(int v) {
        int root = v, child = -1;
        while (v != -1) {
            int p = find_2ecc(par[v]);
            par[v] = child; dsu_cc[v] = root;
            child = v; v = p;
        }
        dsu_cc_size[root] = dsu_cc_size[child];
    }

    void merge_path(int a, int b) {
        ++lca_iter;
        vector<int> path_a, path_b;
        int lca = -1;
        while (lca == -1) {
            if (a != -1) {
                a = find_2ecc(a); path_a.push_back(a);
                if (last_visit[a] == lca_iter) { lca = a; break; }
                last_visit[a] = lca_iter; a = par[a];
            }
            if (b != -1) {
                b = find_2ecc(b); path_b.push_back(b);
                if (last_visit[b] == lca_iter) { lca = b; break; }
                last_visit[b] = lca_iter; b = par[b];
            }
        }
        for (int v : path_a) { dsu_2ecc[v] = lca; if (v == lca) break; --bridges; }
        for (int v : path_b) { dsu_2ecc[v] = lca; if (v == lca) break; --bridges; }
    }

    void add_edge(int a, int b) {
        a = find_2ecc(a); b = find_2ecc(b);
        if (a == b) return;
        int ca = find_cc(a), cb = find_cc(b);
        if (ca != cb) {

```

```

            ++bridges;
            if (dsu_cc_size[ca] > dsu_cc_size[cb]) { swap(a, b);
                swap(ca, cb); }
            make_root(a);
            par[a] = dsu_cc[a] = b;
            dsu_cc_size[cb] += dsu_cc_size[a];
        } else {
            merge_path(a, b);
        }
    }
};

```

- Initialize via DynamicBridge db(n) (0-indexed).
- Call db.add_edge(u, v) online to insert an undirected edge and incrementally update the forest structure.
- Access db.bridges at any execution state to fetch the current total bridge count across all components.

3.6 Dinic Algorithm

Max flow and min-cut algorithm running in $O(V^2E)$ (worst case) and $O(E\sqrt{V})$ on unit networks.

```

const long long INF = 1e18;

struct Dinic {
    struct Edge { int to; long long flow, cap; int rev; };
    int n, source, sink;
    vector<int> level, ptr;
    vector<vector<Edge>> adj;

    Dinic(int n, int source, int sink) : n(n), source(source), sink(
        sink), level(n + 1), ptr(n + 1), adj(n + 1) {}

    void add_edge(int from, int to, long long cap) {
        adj[from].push_back({to, 0, cap, (int)adj[to].size()});
        adj[to].push_back({from, 0, 0, (int)adj[from].size() - 1});
    }

    bool bfs() {
        fill(level.begin(), level.end(), -1);
        level[source] = 0; queue<int> q; q.push(source);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (auto& edge : adj[u]) {
                if (edge.cap - edge.flow > 0 && level[edge.to] ==
                    -1) {
                    level[edge.to] = level[u] + 1; q.push(edge.to);
                }
            }
        }
        return level[sink] != -1;
    }

    long long dfs(int u, long long pushed) {
        if (pushed == 0 || u == sink) return pushed;
        for (int& cid = ptr[u]; cid < adj[u].size(); ++cid) {
            auto& edge = adj[u][cid]; int tr = edge.to;
            if (level[u] + 1 != level[tr] || edge.cap - edge.flow
                == 0) continue;
            long long tr_pushed = dfs(tr, min(pushed, edge.cap -
                edge.flow));
            if (tr_pushed == 0) continue;
            edge.flow += tr_pushed; adj[tr][edge.rev].flow -=
                tr_pushed;
            return tr_pushed;
        }
    }
}

```

```

        return 0;
    }

    long long max_flow() {
        long long flow = 0;
        while (bfs()) {
            fill(ptr.begin(), ptr.end(), 0);
            while (long long pushed = dfs(source, INF)) flow +=
                pushed;
        }
        return flow;
    }

    vector<pair<int, int>> min_cut() {
        vector<pair<int, int>> cuts;
        for (int u = 0; u <= n; ++u) {
            if (level[u] == -1) continue;
            for (auto& edge : adj[u]) {
                if (edge.cap > 0 && level[edge.to] == -1) {
                    cuts.push_back({u, edge.to});
                }
            }
        }
        return cuts;
    }
};

```

- Call max_flow() first to compute the maximum flow from source to sink.
- The array level retains the reachability from source in the residual graph. A node v is on the source side of the cut if level[v] != -1.
- Call min_cut() after max_flow() to get all edges belonging to the minimum cut partition.

3.7 Gomory-Hu Tree

A tree structure representing the minimum cuts (and maximum flows) between all pairs of vertices in an undirected graph using $N - 1$ max-flow calls.

```

struct GomoryHu {
    struct Edge { int u, v; long long w; };
    int n;
    vector<Edge> tree;

    GomoryHu(int _n) : n(_n) {}

    // Requires a Dinic or PushRelabel solver instance "D"
    // initialized with N nodes
    // and the original graph edges added with their respective
    // capacities.
    void build(const vector<pair<pair<int, int>, long long>>& edges) {
        vector<int> p(n, 0);
        for (int i = 1; i < n; ++i) {
            Dinic D(n, i, p[i]); // Replace with your standard
            Dinic(N, S, T)
            for (auto& e : edges) {
                D.add_edge(e.first.first, e.first.second, e.second);
            }
            D.add_edge(e.first.second, e.first.first, e.second);
            // Undirected
        }
        long long flow = D.max_flow();
    }
}

```



```

        tree.push_back({i, p[i], flow});
        for (int j = i + 1; j < n; ++j) {
            if (p[j] == p[i] && D.level[j] != -1) { // j is on
                i's side of the min-cut
                p[j] = i;
            }
        }
    }
};

```

- Initialize via GomoryHu $gh(n)$ (0-indexed). Pass all graph edges to $gh.build(edges)$.
- Assumes your standard Dinic structure can report the min-cut partition via reachable nodes ($D.level[j] \neq -1$ after max_flow).
- The output $tree$ contains exactly $N - 1$ edges. The all-pairs max-flow/min-cut between any u and v equals the minimum edge weight along the path connecting them in this resulting tree.

3.8 Min Cost Max Flow (MCMF)

Finds max flow with min cost in $O(F \cdot E \log V)$ using SPFA (F is the max flow). Negative costs are allowed, but no negative cycles.

```

const long long INF = 1e18;

struct MCMF {
    struct Edge { int to; long long cap, flow, cost; int rev; };
    int n, source, sink;
    vector<vector<Edge>> adj;
    vector<long long> dist;
    vector<int> parent, parent_edge;
    vector<bool> in_queue;

    MCMF(int n, int source, int sink) : n(n), source(source), sink(sink),
        adj(n + 1), dist(n + 1), parent(n + 1), parent_edge(n + 1),
        in_queue(n + 1) {}

    void add_edge(int from, int to, long long cap, long long cost) {
        adj[from].push_back({to, cap, 0, cost, (int)adj[to].size()});
        adj[to].push_back({from, 0, 0, -cost, (int)adj[from].size() - 1});
    }

    bool spfa() {
        fill(dist.begin(), dist.end(), INF); fill(in_queue.begin(),
            in_queue.end(), false);
        queue<int> q; dist[source] = 0; q.push(source); in_queue[source] = true;

        while (!q.empty()) {
            int u = q.front(); q.pop(); in_queue[u] = false;
            for (int i = 0; i < adj[u].size(); i++) {
                auto& edge = adj[u][i];
                if (edge.cap - edge.flow > 0 && dist[u] + edge.cost < dist[edge.to]) {
                    dist[edge.to] = dist[u] + edge.cost; parent[edge.to] = u;
                    parent_edge[edge.to] = i;
                    if (!in_queue[edge.to]) { q.push(edge.to); in_queue[edge.to] = true; }
                }
            }
        }
    }
};

```

```

    }
    return dist[sink] != INF;
}

pair<long long, long long> solve() {
    long long max_flow = 0, min_cost = 0;
    while (spfa()) {
        long long pushed = INF;
        for (int v = sink; v != source; v = parent[v]) {
            pushed = min(pushed, adj[parent[v]][parent_edge[v]].cap - adj[parent[v]][parent_edge[v]].flow);
        }
        for (int v = sink; v != source; v = parent[v]) {
            auto& edge = adj[parent[v]][parent_edge[v]];
            edge.flow += pushed; adj[v][edge.rev].flow -= pushed;
            min_cost += pushed * edge.cost;
        }
        max_flow += pushed;
    }
    return {max_flow, min_cost};
}
};

```

3.9 Flow with Lower Bound

Graph transformation for network flows with lower bounds $[L, C]$.

1. **Transform Graph:** New capacity is $C - L$. Update demands: $\delta[u] - = L$, $\delta[v] + = L$.
2. **Super Source (S') & Sink (T'):** For each vertex u :
 - $\delta[u] > 0$: Add edge (S', u) with capacity $\delta[u]$.
 - $\delta[u] < 0$: Add edge (u, T') with capacity $-\delta[u]$.

Feasible Flow (Circulatory): Run Max Flow $S' \rightarrow T'$.

Feasible iff $max_flow == \sum_{\delta > 0} \delta$. Real flow $= L + current_flow$.

Max/Min Flow with Source (S) and Sink (T):

1. Add edge (T, S) with capacity ∞ . Run Max Flow $S' \rightarrow T'$ to check feasibility.
2. Save $init_flow = flow$ on (T, S) , then remove edge (T, S) .
3. **Max Flow:** Run Max Flow $S \rightarrow T$. Total = $init_flow + new_flow(S, T)$.
4. **Min Flow:** Run Max Flow $T \rightarrow S$. Total = $init_flow - new_flow(T, S)$.

3.10 Heavy-Light Decomposition (HLD)

Path queries/updates in $O(\log^2 N)$ with Segment Tree. Continuous range per heavy path mapped via pos . For edge queries, use $pos[u] + 1$ instead of $pos[u]$ at the final step.

```

struct HLD {
    int n, timer; vector<vector<int>> adj;
    vector<int> parent, depth, heavy, head, pos;

    HLD(int n, const vector<vector<int>>& tree_adj) :
        n(n), timer(0), adj(tree_adj), parent(n + 1), depth(n + 1),
        heavy(n + 1, -1), head(n + 1), pos(n + 1) {}

    int dfs_sz(int u, int p, int d) {
        int size = 1, max_c_size = 0; parent[u] = p; depth[u] = d;
        for (int v : adj[u]) if (v != p) {
            int c_size = dfs_sz(v, u, d + 1); size += c_size;
            if (c_size > max_c_size) { max_c_size = c_size; heavy[u] = v; }
        }
        return size;
    }

    void decompose(int u, int h) {
        head[u] = h; pos[u] = ++timer;
        if (heavy[u] != -1) decompose(heavy[u], h);
        for (int v : adj[u]) if (v != parent[u] && v != heavy[u])
            decompose(v, v);
    }

    void init(int root = 1) { dfs_sz(root, 0, 0); decompose(root, root); }

    template<typename SegTree>
    int query_path(int u, int v, SegTree& seg) {
        int res = 0;
        for (; head[u] != head[v]; v = parent[head[v]]) {
            if (depth[head[u]] > depth[head[v]]) swap(u, v);
            res += seg.query(pos[head[v]], pos[v]);
        }
        if (depth[u] > depth[v]) swap(u, v);
        return res + seg.query(pos[u], pos[v]);
    }

    template<typename SegTree>
    void update_path(int u, int v, int val, SegTree& seg) {
        for (; head[u] != head[v]; v = parent[head[v]]) {
            if (depth[head[u]] > depth[head[v]]) swap(u, v);
            seg.update(pos[head[v]], pos[v], val);
        }
        if (depth[u] > depth[v]) swap(u, v);
        seg.update(pos[u], pos[v], val);
    }
};

```

3.11 Hopcroft-Karp Algorithm

Maximum cardinality matching in bipartite graph in $O(E\sqrt{V})$. Vertices U and V are 1-indexed.

```

const int INF = 1e9;

struct HopcroftKarp {
    int n, m; vector<vector<int>> adj;
    vector<int> pair_u, pair_v, dist;

    HopcroftKarp(int n, int m) : n(n), m(m), adj(n + 1), pair_u(n + 1, 0),
        pair_v(m + 1, 0), dist(n + 1, 0) {}

    void add_edge(int u, int v) { adj[u].push_back(v); }
};

```



```

bool bfs() {
    queue<int> q;
    for (int u = 1; u <= n; u++) {
        if (pair_u[u] == 0) { dist[u] = 0; q.push(u); }
        else dist[u] = INF;
    }
    dist[0] = INF;
    while (!q.empty()) {
        int u = q.front(); q.pop();
        if (dist[u] < dist[0]) {
            for (int v : adj[u]) if (dist[pair_v[v]] == INF) {
                dist[pair_v[v]] = dist[u] + 1; q.push(pair_v[v]);
            }
        }
    }
    return dist[0] != INF;
}

bool dfs(int u) {
    if (!u) return true;
    for (int v : adj[u]) {
        if (dist[pair_v[v]] == dist[u] + 1 && dfs(pair_v[v])) {
            pair_v[v] = u; pair_u[u] = v; return true;
        }
    }
    dist[u] = INF; return false;
}

int max_matching() {
    int matching = 0;
    while (bfs()) {
        for (int u = 1; u <= n; u++) if (pair_u[u] == 0 && dfs(u)) matching++;
    }
    return matching;
}
};

```

3.12 Hungarian Algorithm (Kuhn-Munkres)

Finds the maximum weight (or minimum cost) perfect matching in a complete bipartite graph in $O(V^3)$ time complexity.

```

const long long INF = 1e18;

struct Hungarian {
    int n;
    vector<vector<long long>> cost;
    vector<long long> u, v, dist;
    vector<int> p, way, match;
    vector<bool> used;

    Hungarian(int _n) : n(_n), cost(_n + 1, vector<long long>(_n + 1, 0)), u(_n + 1, 0), v(_n + 1, 0), p(_n + 1, 0), way(_n + 1, 0), match(_n + 1, 0), dist(_n + 1, 0), used(_n + 1, false) {}

    long long solve() {
        for (int i = 1; i <= n; ++i) {
            p[0] = i;
            int j_cur = 0;
            fill(dist.begin(), dist.end(), INF);
            fill(used.begin(), used.end(), false);
            do {
                used[j_cur] = true;
                int i_cur = p[j_cur], j_next = 0;
                long long delta = INF;
                for (int j = 1; j <= n; ++j) {
                    if (!used[j]) {
                        long long cur = cost[i_cur][j] - u[i_cur] -

```

```

                        if (cur < dist[j]) { dist[j] = cur; way[j] = j_cur; }
                        if (dist[j] < delta) { delta = dist[j]; }
                    }
                }
                for (int j = 0; j <= n; ++j) {
                    if (used[j]) { u[p[j]] += delta; v[j] -= delta; }
                    else dist[j] -= delta;
                }
                j_cur = j_next;
            } while (p[j_cur] != 0);
            do {
                int j_prev = way[j_cur];
                p[j_cur] = p[j_prev];
                j_cur = j_prev;
            } while (j_cur != 0);
            for (int j = 1; j <= n; ++j) match[p[j]] = j;
            return -v[0];
        }
    }
};

```

- Initialize via Hungarian hung(n) (1-indexed). Populate hung.cost[i][j] with the cost of matching left node i with right node j .
- Call hung.solve() to find the minimum cost perfect matching. To find the maximum weight instead, negate all elements in the input cost matrix.
- The array hung.match[i] stores the index of the right node matched with left node i .

3.13 Kruskal Reconstruction Tree (KRT)

Max/min edge query on path becomes LCA query. Root of LCA gives node_weight[LCA(u, v)]. Preprocess LCA on adj up to node_cnt ($2N - 1$ nodes).

```

struct KRT {
    int n, node_cnt;
    vector<int> parent, sz, node_weight;
    vector<vector<int>> adj;

    KRT(int n) : n(n), node_cnt(n), parent(2 * n + 1), sz(2 * n + 1, 1), node_weight(2 * n + 1, 0), adj(2 * n + 1) {
        iota(parent.begin(), parent.end(), 0);
    }

    int find_set(int v) {
        return v == parent[v] ? v : parent[v] = find_set(parent[v]);
    }

    void build(vector<tuple<int, int, int>>& edges) {
        sort(edges.begin(), edges.end());
        for (auto [w, u, v] : edges) {
            int root_u = find_set(u), root_v = find_set(v);
            if (root_u != root_v) {
                node_cnt++;
                node_weight[node_cnt] = w;
                parent[root_u] = node_cnt; parent[root_v] = node_cnt;
                adj[node_cnt].push_back(root_u); adj[node_cnt].push_back(root_v);
            }
        }
    }
};

```

```

    }
}
};

```

3.14 Link-Cut Tree

A data structure that maintains a forest of rooted trees under dynamic edge insertions (link) and deletions (cut). All operations run in $O(\log N)$ amortized time using Splay Trees.

```

struct Node {
    int p = 0, ch[2] = {0, 0};
    bool rev = false;
};

vector<Node> t;

void init_lct(int n) {
    t.assign(n + 1, Node());
}

bool is_root(int x) {
    return t[t[x].p].ch[0] != x && t[t[x].p].ch[1] != x;
}

void push(int x) {
    if (x && t[x].rev) {
        swap(t[x].ch[0], t[x].ch[1]);
        if (t[x].ch[0]) t[t[x].ch[0]].rev ^= 1;
        if (t[x].ch[1]) t[t[x].ch[1]].rev ^= 1;
        t[x].rev = false;
    }
}

void push_all(int x) {
    if (!is_root(x)) push_all(t[x].p);
    push(x);
}

void rotate(int x) {
    int y = t[x].p, z = t[y].p;
    int k = (t[y].ch[1] == x);
    if (!is_root(y)) t[z].ch[t[z].ch[1] == y] = x;
    t[x].p = z;
    t[y].ch[k] = t[x].ch[k ^ 1];
    if (t[x].ch[k ^ 1]) t[t[x].ch[k ^ 1]].p = y;
    t[x].ch[k ^ 1] = y;
    t[y].p = x;
}

void splay(int x) {
    push_all(x);
    while (!is_root(x)) {
        int y = t[x].p, z = t[y].p;
        if (!is_root(y)) {
            if ((t[y].ch[1] == x) ^ (t[z].ch[1] == y)) rotate(x);
            else rotate(y);
        }
        rotate(x);
    }
}

void access(int x) {
    for (int t_ch = 0, r = x; r; t_ch = r, r = t[r].p) {
        splay(r);
        t[r].ch[1] = t_ch;
    }
    splay(x);
}

void make_root(int x) {

```

```

    access(x);
    t[x].rev ^= 1;
}

int find_root(int x) {
    access(x);
    while (t[x].ch[0]) {
        push(x);
        x = t[x].ch[0];
    }
    splay(x);
    return x;
}

void link(int x, int y) {
    make_root(x);
    if (find_root(y) != x) t[x].p = y;
}

void cut(int x, int y) {
    make_root(x);
    if (find_root(y) == x && t[y].p == x && !t[y].ch[0]) {
        t[y].p = t[x].ch[1] = 0;
    }
}

```

- Call `init_lct(n)` to allocate memory for n nodes (1-indexed).
- Call `link(x, y)` to add an undirected edge between x and y . It automatically checks connectivity to prevent cycles.
- Call `cut(x, y)` to remove the edge between x and y if it exists.
- Call `make_root(x)` to change the root of the tree containing x to be x .
- Call `find_root(x)` to get the actual component root of x . Useful for dynamic connectivity checks via `find_root(x) == find_root(y)`.

3.15 Boruvka's Algorithm

MST in $O(E \log V)$. Efficient when component-wise cheap edges can be found faster than scanning all edges. Returns -1 if graph is disconnected.

```

struct Boruvka {
    struct Edge { int u, v; long long weight; int id; };
    int n; vector<Edge> edges; vector<int> parent, sz;

    Boruvka(int n) : n(n), parent(n + 1), sz(n + 1, 1) { iota(
        parent.begin(), parent.parent.end(), 0); }

    void add_edge(int u, int v, long long w, int id = 0) { edges.
        push_back({u, v, w, id}); }

    int find_set(int v) { return v == parent[v] ? v : parent[v] =
        find_set(parent[v]); }

    bool union_sets(int a, int b) {
        a = find_set(a); b = find_set(b);
        if (a == b) return false;
        if (sz[a] < sz[b]) swap(a, b);
        parent[b] = a; sz[a] += sz[b]; return true;
    }
}

```

```

}

long long solve(vector<int>& mst_edges) {
    long long total_weight = 0; int num_components = n;
    vector<int> cheapest(n + 1);
    while (num_components > 1) {
        fill(cheapest.begin(), cheapest.end(), -1);
        for (int i = 0; i < edges.size(); i++) {
            int set_u = find_set(edges[i].u), set_v = find_set(
                edges[i].v);
            if (set_u == set_v) continue;
            if (cheapest[set_u] == -1 || edges[i].weight <
                edges[cheapest[set_u]].weight) cheapest[set_u] = i;
            if (cheapest[set_v] == -1 || edges[i].weight <
                edges[cheapest[set_v]].weight) cheapest[set_v] = i;
        }
        bool flag = false;
        for (int i = 1; i <= n; i++) if (cheapest[i] != -1) {
            int idx = cheapest[i];
            if (union_sets(edges[idx].u, edges[idx].v)) {
                total_weight += edges[idx].weight; mst_edges.
                    push_back(edges[idx].id);
                num_components--; flag = true;
            }
        }
        if (!flag) break;
    }
    return (num_components == 1) ? total_weight : -1;
}

```

3.16 Prim's Algorithm

MST in $O(E \log V)$. Grows a single tree component. Returns -1 if graph is disconnected. Populates `mst_edges` with selected edge IDs.

```

const long long INF = 1e18;
struct Edge { int to, weight, id; };

long long prim(int source, int n, const vector<vector<Edge>>& adj,
    vector<int>& mst_edges) {
    long long total_weight = 0; int visited_count = 0;
    vector<bool> visited(n + 1, false);
    vector<long long> min_edge(n + 1, INF);
    vector<int> parent_edge(n + 1, -1);

    priority_queue<pair<long long, int>, vector<pair<long long, int>
        >>, greater<pair<long long, int>>> pq;
    min_edge[source] = 0; pq.push({0, source});

    while (!pq.empty()) {
        auto [w, u] = pq.top(); pq.pop();
        if (visited[u]) continue;

        visited[u] = true; total_weight += w; visited_count++;
        if (parent_edge[u] != -1) mst_edges.push_back(parent_edge[u]);

        for (const auto& edge : adj[u]) {
            if (!visited[edge.to] && edge.weight < min_edge[edge.to]) {
                min_edge[edge.to] = edge.weight; parent_edge[edge.
                    to] = edge.id;
                pq.push({min_edge[edge.to], edge.to});
            }
        }
    }
    return (visited_count == n) ? total_weight : -1;
}

```

3.17 Tarjan Algorithm

Finds Strongly Connected Components (SCC) in directed graphs in $O(V + E)$. Components are labeled 1 to `scc_count` in `scc_id`.

```

struct Tarjan {
    int n, timer, scc_count; vector<vector<int>>& adj;
    vector<int> tin, low, scc_id; vector<bool> in_stack; stack<int>
        st;

    Tarjan(int n, const vector<vector<int>>& adj) :
        n(n), adj(adj), timer(0), scc_count(0), tin(n + 1, -1), low
            (n + 1, -1), scc_id(n + 1, -1), in_stack(n + 1, false) {}

    void dfs(int u) {
        tin[u] = low[u] = ++timer; st.push(u); in_stack[u] = true;
        for (int v : adj[u]) {
            if (tin[v] == -1) {
                dfs(v); low[u] = min(low[u], low[v]);
            } else if (in_stack[v]) {
                low[u] = min(low[u], tin[v]);
            }
        }
        if (low[u] == tin[u]) {
            scc_count++;
            while (true) {
                int v = st.top(); st.pop(); in_stack[v] = false;
                scc_id[v] = scc_count;
                if (u == v) break;
            }
        }
    }

    void run() {
        for (int i = 1; i <= n; i++) if (tin[i] == -1) dfs(i);
    }
};

```

3.18 Virtual Tree (Auxiliary Tree)

Builds a compressed tree of K marked nodes and their LCAs in $O(K \log K)$. Clear `v_adj` via `k_nodes` after use.

```

struct VirtualTree {
    LCA &lca; vector<vector<int>> v_adj;
    VirtualTree(LCA &lca) : lca(lca), v_adj(lca.n + 1) {}

    void build(vector<int> &k_nodes) {
        auto cmp = [&](int a, int b) { return lca.tin[a] < lca.tin[
            b]; };
        sort(k_nodes.begin(), k_nodes.end(), cmp);

        int m = k_nodes.size();
        for (int i = 0; i < m - 1; i++) k_nodes.push_back(lca.query
            (k_nodes[i], k_nodes[i + 1]));
        sort(k_nodes.begin(), k_nodes.end(), cmp);
        k_nodes.erase(unique(k_nodes.begin(), k_nodes.end()),
            k_nodes.end());

        for (int u : k_nodes) v_adj[u].clear();
        vector<int> st;
        for (int u : k_nodes) {
            while (!st.empty() && !lca.is_ancestor(st.back(), u))
                st.pop_back();
            if (!st.empty()) v_adj[st.back()].push_back(u);
            st.push_back(u);
        }
    }
};

```

4 Math

4.1 Chinese Remainder Theorem (CRT)

Solves $x \equiv r_i \pmod{m_i}$ for non-coprime moduli. Returns $\{\text{ans}, \text{lcm}\}$, or $\{-1, -1\}$ if impossible.

```
long long extgcd(long long a, long long b, long long &x, long long &y) {
    if (!b) { x = 1; y = 0; return a; }
    long long x1, y1, d = extgcd(b, a % b, x1, y1);
    x = y1; y = x1 - y1 * (a / b); return d;
}

pair<long long, long long> crt_two(long long r1, long long m1, long long r2, long long m2) {
    long long x, y, g = extgcd(m1, m2, x, y);
    if ((r2 - r1) % g != 0) return {-1, -1};
    long long lcm = m1 / g * m2, mod = m2 / g;
    long long x0 = ((x % mod) + mod) % mod;
    long long ans = (r1 + (__int128)((r2 - r1) / g) * x0 % lcm * m1) % lcm;
    return {ans < 0 ? ans + lcm : ans, lcm};
}

pair<long long, long long> crt(const vector<pair<long long, long long>>& eq) {
    if (eq.empty()) return {0, 1};
    long long r_ans = eq[0].first, m_ans = eq[0].second;
    for (size_t i = 1; i < eq.size(); i++) {
        auto [r, m] = crt_two(r_ans, m_ans, eq[i].first, eq[i].second);
        if (m == -1) return {-1, -1};
        r_ans = r; m_ans = m;
    }
    return {r_ans, m_ans};
}
```

Usage guide for $x \equiv r_i \pmod{m_i}$:

- Pass equations as $\{r_i, m_i\}$ into `vector<pair<long long, long long>> eq`.
- Call `auto [ans, lcm] = crt(eq);`. If `lcm == -1`, no solution exists.
- Safe guard: Ensure $0 \leq r_i < m_i$ before calling to avoid incorrect `r2 - r1` signs inside.

4.2 Euler's Totient Function (Phi)

Single query in $O(\sqrt{N})$. Sieve precomputation up to N in $O(N \log \log N)$.

```
long long phi_single(long long n) {
    long long res = n;
    for (long long i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            while (n % i == 0) n /= i;
            res -= res / i;
        }
    }
    if (n > 1) res -= res / n;
    return res;
}

vector<int> phi_sieve(int n) {
    vector<int> phi(n + 1);
    for (int i = 0; i <= n; i++) phi[i] = i;
    for (int i = 2; i <= n; i++) if (phi[i] == i) {
```

```
        for (int j = i; j <= n; j += i) phi[j] -= phi[j] / i;
    }
    return phi;
}
```

4.3 Extended Euclidean

Computes $\gcd(a, b)$ and finds x, y such that $ax + by = \gcd(a, b)$ in $O(\log(\min(a, b)))$. Modular inverse of $a \pmod{m}$ is $(x \% m + m) \% m$ if $\gcd == 1$.

```
long long extgcd(long long a, long long b, long long &x, long long &y) {
    if (!b) { x = 1; y = 0; return a; }
    long long x1, y1, d = extgcd(b, a % b, x1, y1);
    x = y1; y = x1 - y1 * (a / b); return d;
}
```

4.4 FFT & NTT

Polynomial multiplication in $O(N \log N)$. Use `FFT` for high-precision floating-point coefficients and `NTT` for exact integer multiplication under a large prime modulus (typically 998244353).

```
const double PI = acos(-1);
struct Complex {
    double r, i;
    Complex(double r = 0, double i = 0) : r(r), i(i) {}
    Complex operator+(const Complex& o) const { return {r + o.r, i + o.i}; }
    Complex operator-(const Complex& o) const { return {r - o.r, i - o.i}; }
    Complex operator*(const Complex& o) const { return {r * o.r - i * o.i, r * o.i + i * o.r}; }
};

void fft(vector<Complex>& a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len <= n; len <= 1) {
        double ang = 2 * PI / len * (invert ? -1 : 1);
        Complex wlen(cos(ang), sin(ang));
        for (int i = 0; i < n; i += len) {
            Complex w(1);
            for (int j = 0; j < len / 2; j++) {
                Complex u = a[i + j], v = a[i + j + len / 2] * w;
                a[i + j] = u + v;
                a[i + j + len / 2] = u - v;
                w = w * wlen;
            }
        }
    }
    if (invert) {
        for (auto& x : a) x.r /= n;
    }
}

vector<long long> multiply_fft(const vector<int>& a, const vector<int>& b) {
    vector<Complex> fa(a.begin(), a.end()), fb(b.begin(), b.end());
    int n = 1; while (n < a.size() + b.size()) n <= 1;
    fa.resize(n); fb.resize(n);
    fft(fa, false); fft(fb, false);
    for (int i = 0; i < n; i++) fa[i] = fa[i] * fb[i];
```

```
fft(fa, true);
vector<long long> res(n);
for (int i = 0; i < n; i++) res[i] = round(fa[i].r);
return res;
}
```

```
const int MOD = 998244353;
const int G = 3; // Primitive root for 998244353

long long power(long long base, long long p) {
    long long res = 1; base %= MOD;
    while (p > 0) {
        if (p & 1) res = (res * base) % MOD;
        base = (base * base) % MOD;
        p >>= 1;
    }
    return res;
}

void ntt(vector<long long>& a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len <= n; len <= 1) {
        long long wlen = power(G, (MOD - 1) / len);
        if (invert) wlen = power(wlen, MOD - 2);
        for (int i = 0; i < n; i += len) {
            long long w = 1;
            for (int j = 0; j < len / 2; j++) {
                long long u = a[i + j], v = (a[i + j + len / 2] * w) % MOD;
                a[i + j] = (u + v < MOD ? u + v : u + v - MOD);
                a[i + j + len / 2] = (u - v >= 0 ? u - v : u - v + MOD);
                w = (w * wlen) % MOD;
            }
        }
    }
    if (invert) {
        long long n_inv = power(n, MOD - 2);
        for (auto& x : a) x = (x * n_inv) % MOD;
    }
}

vector<long long> multiply_ntt(const vector<long long>& a, const vector<long long>& b) {
    vector<long long> fa(a.begin(), a.end()), fb(b.begin(), b.end());
    int n = 1; while (n < a.size() + b.size()) n <= 1;
    fa.resize(n); fb.resize(n);
    ntt(fa, false); ntt(fb, false);
    for (int i = 0; i < n; i++) fa[i] = (fa[i] * fb[i]) % MOD;
    ntt(fa, true);
    return fa;
}
```

To multiply two polynomials $A(x)$ and $B(x)$, pass their coefficients to `multiply_fft` or `multiply_ntt`. For example:

```
// Multiply (1 + 2x) * (2 + 3x) -> 2 + 7x + 6x^2
vector<int> a = {1, 2};
vector<int> b = {2, 3};
vector<long long> c = multiply_fft(a, b);
// c contains: {2, 7, 6, 0, ...}
```

4.5 Matrix Exponentiation Template

Solves linear recurrence relations and DP optimizations in $O(K^3 \log N)$. Handles modular addition, subtraction, multiplication, and fast power.

```
const int MOD = 1e9 + 7;

struct Matrix {
    int r, c; vector<vector<long long>> mat;

    Matrix(int r, int c, bool identity = false) : r(r), c(c), mat(r, vector<long long>(c, 0)) {
        if (identity) for (int i = 0; i < min(r, c); i++) mat[i][i] = 1;
    }

    Matrix operator+(const Matrix& o) const {
        Matrix res(r, c);
        for (int i = 0; i < r; i++) for (int j = 0; j < c; j++)
            res.mat[i][j] = (mat[i][j] + o.mat[i][j]) % MOD;
        return res;
    }

    Matrix operator-(const Matrix& o) const {
        Matrix res(r, c);
        for (int i = 0; i < r; i++) for (int j = 0; j < c; j++)
            res.mat[i][j] = (mat[i][j] - o.mat[i][j] + MOD) % MOD;
        return res;
    }

    Matrix operator*(const Matrix& o) const {
        Matrix res(r, o.c);
        for (int i = 0; i < r; i++) for (int k = 0; k < c; k++) for (int j = 0; j < o.c; j++)
            res.mat[i][j] = (res.mat[i][j] + mat[i][k] * o.mat[k][j]) % MOD;
        return res;
    }

    Matrix power(long long p) const {
        Matrix res(r, r, true), base = *this;
        while (p > 0) {
            if (p & 1) res = res * base;
            base = base * base; p >>= 1;
        }
        return res;
    }
};
```

Fibonacci F_n example ($T^{n-1} \cdot F_1$):

```
Matrix T(2, 2); T.mat = {{1, 1}, {1, 0}};
Matrix F(2, 1); F.mat = {{1}, {0}}; // F[1]=1, F[0]=0
```

```
Matrix Ans = T.power(n - 1) * F;
long long fn = Ans.mat[0][0];
```

4.6 Miller-Rabin Primality Test & Pollard's Rho

Deterministic 64-bit primality check in $O(\log N)$ and factorization in $O(N^{1/4})$.

```
typedef unsigned long long ull;

ull mul_mod(ull a, ull b, ull m) {
    long long res = a * b - m * (ull)((long double)a * b / m);
    return (res < 0 ? res + m : res) % m;
}
```

```
ull power(ull base, ull p, ull mod) {
    ull res = 1; base %= mod;
    while (p > 0) {
        if (p & 1) res = mul_mod(res, base, mod);
        base = mul_mod(base, base, mod); p >>= 1;
    }
    return res;
}

bool miller_rabin(ull n) {
    if (n < 2) return false;
    if (n == 2 || n == 3) return true;
    if (n % 2 == 0) return false;
    ull d = n - 1; int s = 0;
    while (d % 2 == 0) { d /= 2; s++; }
    static const ull bases[] = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37};
    for (ull a : bases) {
        if (n <= a) break;
        ull x = power(a, d, n);
        if (x == 1 || x == n - 1) continue;
        bool composite = true;
        for (int r = 1; r < s; r++) {
            x = mul_mod(x, x, n);
            if (x == n - 1) { composite = false; break; }
        }
        if (composite) return false;
    }
    return true;
}

ull pollard_rho(ull n) {
    if (n % 2 == 0) return 2;
    if (miller_rabin(n)) return n;
    ull x = 2, y = 2, d = 1, c = 1;
    auto f = [&](ull x, ull n, ull c) { return (mul_mod(x, x, n) + c) % n; };
    while (d == 1) {
        x = f(x, n, c); y = f(f(y, n, c), n, c);
        d = std::gcd(x > y ? x - y : y - x, n);
        if (d == n) { x = rand() % (n - 2) + 2; y = x; c = rand() % (n - 1) + 1; d = 1; }
    }
    return d;
}

void factorize(ull n, vector<ull>& factors) {
    if (n == 1) return;
    if (miller_rabin(n)) { factors.push_back(n); return; }
    ull p = pollard_rho(n); factorize(p, factors); factorize(n / p, factors);
}
```

4.7 Binary GCD (Stein's Algorithm)

```
long long fast_gcd(long long a, long long b) {
    if (!a || !b) return a ^ b;
    int shift = __builtin_ctzll(a | b);
    a >>= __builtin_ctzll(a);
    while (b) {
        b >>= __builtin_ctzll(b);
        if (a > b) swap(a, b);
        b -= a;
    }
    return a << shift;
}
```

4.8 Stern-Brocot Tree

A data structure that generates all positive irreducible fractions exactly once by maintaining lower and upper bounds.

```
struct Fraction {
    long long p, q;
    bool operator==(const Fraction& o) const { return p * o.q == q * o.p; }
    bool operator<(const Fraction& o) const { return p * o.q < q * o.p; }
};

struct SternBrocot {
    Fraction mediant(Fraction l, Fraction r) {
        return {l.p + r.p, l.q + r.q};
    }

    string find_path(Fraction target, int max_depth = 60) {
        Fraction L = {0, 1}, R = {1, 0};
        string path = "";
        for (int i = 0; i < max_depth; ++i) {
            Fraction M = mediant(L, R);
            if (target == M) break;
            if (target < M) {
                path += 'L';
                R = M;
            } else {
                path += 'R';
                L = M;
            }
        }
        return path;
    }
};
```

- Call `mediant(l, r)` to get the fraction between bounds l and r .
- Call `find_path(target)` to trace the path from the root to the target.
- Every generated fraction satisfies $\gcd(p, q) = 1$ automatically.

4.9 Tonelli-Shanks Algorithm

Solves the modular congruence $x^2 \equiv n \pmod{p}$ for a prime p in $O(\log^2 p)$ time complexity. It finds the modular square root if it exists.

```
long long power(long long base, long long p, long long mod) {
    long long res = 1; base %= mod;
    while (p > 0) {
        if (p & 1) res = (__int128)res * base % mod;
        base = (__int128)base * base % mod;
        p >>= 1;
    }
    return res;
}

long long tonelli_shanks(long long n, long long p) {
    if (p == 2) return n & 1;
    if (power(n, (p - 1) / 2, p) != 1) return -1; // No solution (Legendre symbol)
    if (p % 4 == 3) return power(n, (p + 1) / 4, p);

    long long q = p - 1, s = 0;
    while (q % 2 == 0) { q /= 2; s++; }

    long long z = 2;
    while (power(z, (p - 1) / 2, p) == 1) z++;

    long long m = s, c = power(z, q, p), t = power(n, q, p), r = power(n, (q + 1) / 2, p);
```

```

while (t != 1) {
    long long t2k = t, k = 0;
    for (k = 1; k < m; k++) {
        t2k = (__int128)t2k * t2k % p;
        if (t2k == 1) break;
    }
    if (k == m) return -1;
    long long b = power(c, 1LL << (m - k - 1), p);
    r = (__int128)r * b % p;
    c = (__int128)b * b % p;
    t = (__int128)t * c % p;
    m = k;
}
return r;
}

```

Call `tonelli_shanks(n, p)` to find a value x such that $x^2 \equiv n \pmod{p}$.

- Returns a valid solution $x \in [0, p - 1]$ if it exists.
- Returns -1 if n is a quadratic non-residue modulo p (no solution exists).
- Note that if x is a solution, $p - x$ is also a valid solution.

5 Strings

5.1 Aho-Corasick Automaton

Multi-pattern matching in $O(\sum |pattern| + |text|)$. Converts Trie to DFA via transition table optimization.

```

struct AhoCorasick {
    static const int ALPHABET_SIZE = 26;
    struct Node {
        int next[ALPHABET_SIZE], link = -1, exit_link = -1;
        vector<int> ids;
        Node() { memset(next, -1, sizeof(next)); }
    };

    vector<Node> trie;
    AhoCorasick() { trie.push_back(Node()); }

    void insert(const string& s, int id) {
        int u = 0;
        for (char c : s) {
            int idx = c - 'a';
            if (trie[u].next[idx] == -1) {
                trie[u].next[idx] = trie.size(); trie.push_back(
                    Node());
                u = trie[u].next[idx];
            }
            trie[u].ids.push_back(id);
        }
    }

    void build() {
        queue<int> q;
        for (int i = 0; i < ALPHABET_SIZE; i++) {
            if (trie[0].next[i] != -1) { trie[trie[0].next[i]].link = 0; q.push(trie[0].next[i]); }
            else trie[0].next[i] = 0;
        }
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (int i = 0; i < ALPHABET_SIZE; i++) {
                int v = trie[u].next[i];
                if (v != -1) {

```

```

                trie[v].link = trie[trie[u].link].next[i];
                int slink = trie[v].link;
                trie[v].exit_link = !trie[slink].ids.empty() ?
                    slink : trie[slink].exit_link;
                q.push(v);
            } else {
                trie[u].next[i] = trie[trie[u].link].next[i];
            }
        }
    }
};

```

5.2 KMP Algorithm

```

vector<int> prefix_function(string s) {
    int n = s.length(); vector<int> pi(n, 0);
    for (int i = 1; i < n; i++) {
        int j = pi[i - 1];
        while (j > 0 && s[i] != s[j]) j = pi[j - 1];
        if (s[i] == s[j]) j++;
        pi[i] = j;
    }
    return pi;
}

vector<int> kmp_search(string text, string pattern) {
    string s = pattern + "#" + text;
    int n = s.length(), m = pattern.length();
    vector<int> pi = prefix_function(s, matches);
    for (int i = m + 1; i < n; i++) {
        if (pi[i] == m) matches.push_back(i - 2 * m);
    }
    return matches;
}

```

5.3 Manacher's Algorithm

Computes palindromic radii for all positions in $O(N)$. Max palindrome length at i is $p[i]$.

```

vector<int> manacher(string s) {
    string t = "#";
    for (char c : s) t += c, t += "#";
    int n = t.length(); vector<int> p(n, 0);
    int c = 0, r = 0;

    for (int i = 0; i < n; i++) {
        int mirror = 2 * c - i;
        if (i < r) p[i] = min(r - i, p[mirror]);
        while (i + p[i] + 1 < n && i - p[i] - 1 >= 0 && t[i + p[i] + 1] == t[i - p[i] - 1]) p[i]++;
        if (i + p[i] > r) { c = i; r = i + p[i]; }
    }
    return p;
}

```

5.4 Palindrome Tree (Eertree)

A specialized tree structure that processes characters online to maintain all unique palindromic substrings of a string in $O(N)$ time and space complexity.

```

struct PalindromeTree {
    static const int ALPHABET_SIZE = 26;
    struct Node {
        int next[ALPHABET_SIZE];

```

```

        int len;
        int link;
        int cnt; // Frequency of the palindrome substring
    };

    string s;
    vector<Node> tree;
    int sz, last;

    PalindromeTree() {
        s = "";
        // Node 1: Imaginary root of odd length (-1)
        // Node 2: Root of even length (0)
        tree.push_back({}); // Dummy at index 0
        tree.push_back({}); tree[1].len = -1; tree[1].link = 1;
        tree.push_back({}); tree[2].len = 0; tree[2].link = 1;
        memset(tree[1].next, 0, sizeof(tree[1].next));
        memset(tree[2].next, 0, sizeof(tree[2].next));
        sz = 2; last = 2;
    }

    int get_link(int u) {
        int curr_pos = s.length() - 1;
        while (true) {
            int prev_pos = curr_pos - 1 - tree[u].len;
            if (prev_pos >= 0 && s[prev_pos] == s[curr_pos]) return
                u;
            u = tree[u].link;
        }
    }

    void insert(char ch) {
        s += ch;
        int c = ch - 'a';
        int curr = get_link(last);

        if (tree[curr].next[c]) {
            last = tree[curr].next[c];
            tree[last].cnt++;
            return;
        }

        sz++;
        last = sz;
        tree.push_back({});
        memset(tree[sz].next, 0, sizeof(tree[sz].next));

        tree[sz].len = tree[curr].len + 2;
        tree[curr].next[c] = sz;

        if (tree[sz].len == 1) {
            tree[sz].link = 2;
            tree[sz].cnt = 1;
            return;
        }

        int link_node = get_link(tree[curr].link);
        tree[sz].link = tree[link_node].next[c];
        tree[sz].cnt = 1;
    }

    void pull_frequencies() {
        // Propagate counts from longer palindromes to their suffix
        links
        for (int i = sz; i > 2; i--) {
            tree[tree[i].link].cnt += tree[i].cnt;
        }
    }
};

```

Initialize via `PalindromeTree pt` and append characters one by one using `pt.insert(c)`. Call `pt.pull_frequencies()` at the end to get the exact total occurrence count of every unique palindromic

substring inside `tree[i].cnt`.

5.5 String Double Hashing

Anti-hash-kill rolling hash with random bases. Input range $[l, r]$ for `get(l, r)` is 0-indexed.

```
struct StringHash {
    long long MOD1 = 1e9 + 7, MOD2 = 1e9 + 9, BASE1, BASE2;
    vector<long long> h1, h2, p1, p2;

    StringHash(const string& s) {
        BASE1 = (rng() % (MOD1 - 300) + 300) | 1;
        BASE2 = (rng() % (MOD2 - 300) + 300) | 1;
        int n = s.size();
        h1.resize(n + 1, 0); h2.resize(n + 1, 0);
        p1.resize(n + 1, 1); p2.resize(n + 1, 1);
        for (int i = 0; i < n; i++) {
            h1[i + 1] = (h1[i] * BASE1 + s[i]) % MOD1;
            h2[i + 1] = (h2[i] * BASE2 + s[i]) % MOD2;
            p1[i + 1] = (p1[i] * BASE1) % MOD1;
            p2[i + 1] = (p2[i] * BASE2) % MOD2;
        }
    }

    pair<long long, long long> get(int l, int r) {
        l++; r++;
        long long res1 = (h1[r] - h1[l - 1] * p1[r - l + 1]) % MOD1;
        long long res2 = (h2[r] - h2[l - 1] * p2[r - l + 1]) % MOD2;
        return {res1 < 0 ? res1 + MOD1 : res1, res2 < 0 ? res2 + MOD2 : res2};
    }
};
```

5.6 Suffix Array

Constructs SA and LCP in $O(N \log N)$. Appends sentinel '\$' internally. `sa[i]` is the first real suffix, `lcp[i]` stores LCP between `sa[i]` and `sa[i-1]`.

```
struct SuffixArray {
    string s; int n; vector<int> sa, rank, lcp;
    SuffixArray(string s) : s(s + "$"), n(s.length() + 1) { sa.resize(n); rank.resize(n); build_sa(); build_lcp(); }

    void build_sa() {
        vector<int> p(n), c(n), cnt(max(256, n), 0), pn(n), cn(n);
        for (int i = 0; i < n; i++) cnt[s[i]]++;
        for (int i = 1; i < 256; i++) cnt[i] += cnt[i - 1];
        for (int i = 0; i < n; i++) p[--cnt[s[i]]] = i;
        c[p[0]] = 0; int classes = 1;
        for (int i = 1; i < n; i++) {
            if (s[p[i]] != s[p[i - 1]]) classes++;
            c[p[i]] = classes - 1;
        }
        for (int h = 0; (1 << h) < n; ++h) {
            for (int i = 0; i < n; i++) pn[i] = (p[i] - (1 << h) + n) % n;
            fill(cnt.begin(), cnt.begin() + classes, 0);
            for (int i = 0; i < n; i++) cnt[c[pn[i]]]++;
            for (int i = 1; i < classes; i++) cnt[i] += cnt[i - 1];
            for (int i = n - 1; i >= 0; i--) p[--cnt[c[pn[i]]]] = pn[i];
            cn[p[0]] = 0; classes = 1;
            for (int i = 1; i < n; i++) {
                if (c[p[i]] != c[p[i - 1]] || c[p[i] + (1 << h)] % n != c[p[i - 1] + (1 << h)] % n) classes++;
                cn[p[i]] = classes - 1;
            }
        }
    }
};
```

```
        c = cn;
    }
    sa = p; for (int i = 0; i < n; i++) rank[sa[i]] = i;
}

void build_lcp() {
    lcp.assign(n, 0); int k = 0;
    for (int i = 0; i < n - 1; i++) {
        int pi = rank[i], j = sa[pi - 1];
        while (s[i + k] == s[j + k]) k++;
        lcp[pi] = k; if (k) k--;
    }
}
};
```

5.7 Suffix Automaton (SAM)

A directed acyclic word graph that represents all suffixes of a string in $O(N)$ time and space. Each state represents a set of substrings that share the same end-positions (endpos).

```
struct State {
    int len, link;
    int next[26];
};

const int MAXLEN = 100005;
State st[MAXLEN * 2];
int sz, last;

void sam_init() {
    st[0].len = 0;
    st[0].link = -1;
    memset(st[0].next, -1, sizeof(st[0].next));
    sz = 1;
    last = 0;
}

void sam_extend(char c) {
    int cur = sz++;
    st[cur].len = st[last].len + 1;
    memset(st[cur].next, -1, sizeof(st[cur].next));

    int p = last;
    while (p != -1 && st[p].next[c - 'a'] == -1) {
        st[p].next[c - 'a'] = cur;
        p = st[p].link;
    }

    if (p == -1) {
        st[cur].link = 0;
    } else {
        int q = st[p].next[c - 'a'];
        if (st[p].len + 1 == st[q].len) {
            st[cur].link = q;
        } else {
            int clone = sz++;
            st[clone].len = st[p].len + 1;
            memcpy(st[clone].next, st[q].next, sizeof(st[q].next));
            st[clone].link = st[q].link;
            while (p != -1 && st[p].next[c - 'a'] == q) {
                st[p].next[c - 'a'] = clone;
                p = st[p].link;
            }
            st[q].link = st[cur].link = clone;
        }
    }
    last = cur;
}

};
```

- Call `sam_init()` once before building the automaton.

- Call `sam_extend(c)` sequentially for each character in the string to construct the SAM online.
- The total number of states in the automaton is at most $2N - 1$. Ensure `MAXLEN` is at least the maximum string length.
- Useful for complex string operations such as substring counting, longest common substring of multiple strings, and finding the lexicographically k -th substring.

5.8 Trie

Array-based Trie acting as a multiset. Allows duplicate words, $O(L)$ insertion, search, and deletion.

```
struct Trie {
    static const int ALPHABET_SIZE = 26;
    struct Node {
        int next[ALPHABET_SIZE], cnt = 0, is_end = 0;
        Node() { memset(next, -1, sizeof(next)); }
    };

    vector<Node> trie;
    Trie() { trie.push_back(Node()); }

    void insert(const string& s) {
        int u = 0; trie[u].cnt++;
        for (char c : s) {
            int idx = c - 'a';
            if (trie[u].next[idx] == -1) {
                trie[u].next[idx] = trie.size(); trie.push_back(Node());
            }
            u = trie[u].next[idx]; trie[u].cnt++;
        }
        trie[u].is_end++;
    }

    bool search(const string& s) {
        int u = 0;
        for (char c : s) {
            int idx = c - 'a';
            if (trie[u].next[idx] == -1 || trie[trie[u].next[idx]].cnt == 0) return false;
            u = trie[u].next[idx];
        }
        return trie[u].is_end > 0;
    }

    void remove(const string& s) {
        if (!search(s)) return;
        int u = 0; trie[u].cnt--;
        for (char c : s) {
            int idx = c - 'a', nxt = trie[u].next[idx];
            trie[nxt].cnt--; u = nxt;
        }
        trie[u].is_end--;
    }
};
```

5.9 Z-Algorithm

Computes Z-array in $O(N)$ where $Z[i]$ is the LCP length of s and its suffix starting at i .


```
vector<int> z_function(string s) {
    int n = s.length(); vector<int> z(n, 0);
    int l = 0, r = 0;
    for (int i = 1; i < n; i++) {
        if (i <= r) z[i] = min(r - i + 1, z[i - 1]);
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) z[i]++;
        if (i + z[i] - 1 > r) { l = i; r = i + z[i] - 1; }
    }
    return z;
}

vector<int> z_search(string text, string pattern) {
    string s = pattern + "#" + text; int m = pattern.length();
    vector<int> z = z_function(s), matches;
    for (int i = m + 1; i < s.length(); i++) {
        if (z[i] == m) matches.push_back(i - m - 1);
    }
    return matches;
}
```

6 Bit Manipulation

6.1 Binary Trie (Bit Trie)

$O(B)$ max XOR query on 32-bit integers. Acting as a multiset using node counters.

```
struct BitTrie {
    static const int BITS = 30;
    struct Node {
        int next[2] = {0, 0}, cnt = 0;
    };
    vector<Node> trie;
    BitTrie() { trie.push_back(Node()); }

    void insert(int x) {
        int u = 0; trie[u].cnt++;
        for (int i = BITS; i >= 0; i--) {
            int bit = (x >> i) & 1;
            if (!trie[u].next[bit]) { trie[u].next[bit] = trie.size();
                trie.push_back(Node()); }
            u = trie[u].next[bit]; trie[u].cnt++;
        }
    }

    void remove(int x) {
        int u = 0; trie[u].cnt--;
        for (int i = BITS; i >= 0; i--) {
            u = trie[u].next[(x >> i) & 1]; trie[u].cnt--;
        }
    }

    int max_xor(int x) {
        int u = 0, res = 0;
        for (int i = BITS; i >= 0; i--) {
            int bit = (x >> i) & 1, toggle = bit ^ 1;
            if (trie[u].next[toggle] && trie[trie[u].next[toggle]].cnt > 0) {
                res |= (1 << i); u = trie[u].next[toggle];
            } else {
                u = trie[u].next[bit];
            }
        }
        return res;
    }
};
```

6.2 Bitwise Hacks

High-performance bitwise manipulations. $O(1)$ submask traversal and hardware-accelerated queries.

```
// 1. Core Bit Operations
x | (1LL << i) // Set i-th bit to 1
x & ~(1LL << i) // Set i-th bit to 0
x ^ (1LL << i) // Toggle i-th bit
(x >> i) & 1LL // Check if i-th bit is 1
long long lsb = x & -x; // Isolate lowest set bit (Fenwick)
long long clear_lsb = x & (x - 1); // Clear lowest set bit

// 2. Bulk & Range Bit Manipulations (k bits, Range [l, r])
(1LL << k) - 1 // Mask of k lowest bits (000...111)
x ^ ((1LL << k) - 1) // Toggle k lowest bits of x
(1LL << (r - l + 1)) - 1 << l // Mask for range [l, r] (inclusive)
x ^ (((1LL << (r - l + 1)) - 1) << l) // Toggle all bits in range [l, r]
x | (x + 1) // Set the lowest unset (0) bit to 1

// 3. Submask Traversal
for (int sub = mask; sub > 0; sub = (sub - 1) & mask) {} // Iterate all submasks

// 4. Built-ins & Helper Functions
__builtin_popcountll(x) // Count set bits
__builtin_ctzll(x) // Count trailing zeros (x > 0)
__builtin_clzll(x) // Count leading zeros (x > 0)
_lg(x) // Fast Floor(log2(x)) -> Sparse Table query
long long next_p2(long long x) { return x <= 1 ? 1 : 1LL << (64 - __builtin_clzll(x - 1)); } // Next power of 2
long long safe_mod_mul(unsigned long long a, unsigned long long b, unsigned long long m) { // O(1) mul modulo without overflow
    long long res; return __builtin_mul_overflow(a, b, &res) ? (long long)((unsigned __int128)a * b) % m : res % m;
}
```

6.3 XOR Basis

Vector space for maximum XOR subset queries in $O(B)$ time.

```
struct XorBasis {
    static const int BITS = 60;
    long long basis[BITS]; int sz = 0;
    XorBasis() { memset(basis, 0, sizeof(basis)); }

    bool insert(long long mask) {
        for (int i = BITS - 1; i >= 0; i--) if ((mask >> i) & 1) {
            if (!basis[i]) { basis[i] = mask; sz++; return true; }
            mask ^= basis[i];
        }
        return false;
    }

    long long max_xor() {
        long long res = 0;
        for (int i = BITS - 1; i >= 0; i--) if ((res ^ basis[i]) > res) res ^= basis[i];
        return res;
    }
};
```

7 DPs

7.1 Aliens Trick (WQS Binary Search)

Convex/concave function DP optimization from $O(N \cdot K)$ to $O(N \log(\text{Max } C))$. Binary search penalty λ for choosing

exactly K items.

```
pair<long long, int> check(long long lambda) {
    vector<pair<long long, int>> dp(n + 1, {1e18, 0}); dp[0] = {0, 0};
    for (int i = 1; i <= n; i++) {
        dp[i] = dp[i - 1];
        long long cost = dp[i - 1].first + a[i] - lambda;
        if (cost < dp[i].first) dp[i] = {cost, dp[i - 1].second + 1};
    }
    return dp[n];
}

long long solve_alien() {
    long long l = 0, r = 1e12, ans = -1;
    while (l <= r) {
        long long mid = (l + r) >> 1; auto [cost, count] = check(mid);
        if (count >= k) { ans = cost + k * mid; r = mid - 1; }
        else l = mid + 1;
    }
    return ans;
}
```

7.2 Bitmask DP for Set Partitioning

Solves the optimal grouping problem (CSES 1653 "Elevator Rides") for N items with a maximum capacity X per group in $O(N \cdot 2^N)$ time complexity.

```
pair<int, long long> solve_elevator(int n, long long max_weight, const vector<long long>& w) {
    vector<pair<int, long long>> dp(1 << n);
    dp[0] = {1, 0};

    for (int mask = 1; mask < (1 << n); ++mask) {
        dp[mask] = {n + 1, 0};
        for (int i = 0; i < n; ++i) {
            if (mask & (1 << i)) {
                auto prev = dp[mask ^ (1 << i)];
                if (prev.second + w[i] <= max_weight) {
                    prev.second += w[i];
                } else {
                    prev.first++;
                    prev.second = w[i];
                }
                dp[mask] = min(dp[mask], prev);
            }
        }
    }
    return dp[(1 << n) - 1];
}
```

- $\text{dp}[\text{mask}]$ stores a pair representing the minimum number of groups and the weight of the last group for a subset of items.
- Pairs are compared lexicographically: it minimizes the number of groups first, then minimizes the current group's weight.

7.3 Divide & Conquer Optimization

Reduces DP from $O(N^2)$ to $O(N \log N)$ when transition satisfies monotonicity: $\text{opt}[i][j] \leq \text{opt}[i][j + 1]$.


```

long long cost(int k, int j) { return 0; }

void compute(int l, int r, int opt_l, int opt_r) {
    if (l > r) return;
    int mid = (l + r) >> 1, best_k = -1; dp_curr[mid] = 1e18;

    for (int k = opt_l; k <= min(mid, opt_r); k++) {
        long long current_cost = dp_before[k] + cost(k, mid);
        if (current_cost < dp_curr[mid]) { dp_curr[mid] =
            current_cost; best_k = k; }
    }
    compute(l, mid - 1, opt_l, best_k); compute(mid + 1, r, best_k,
        opt_r);
}

void solve_dp() {
    dp_before.assign(n + 1, 0); dp_curr.assign(n + 1, 0);
    for (int j = 1; j <= n; j++) dp_before[j] = cost(0, j);
    for (int i = 2; i <= g; i++) { compute(1, n, 0, n); dp_before =
        dp_curr; }
}

```

7.4 Knuth Optimization

Reduces DP from $O(N^3)$ to $O(N^2)$ when $opt[i][j - 1] \leq opt[i][j] \leq opt[i + 1][j]$. Cost function must satisfy quadrangle inequality and monotonicity.

```

long long cost(int i, int j) { return 0; }

void solve_dp() {
    dp.assign(n + 1, vector<long long>(n + 1, 0)); opt.assign(n +
        1, vector<int>(n + 1, 0));
    for (int i = 1; i <= n; i++) opt[i][i] = i;

    for (int len = 2; len <= n; len++) {
        for (int i = 1; i + len - 1 <= n; i++) {
            int j = i + len - 1; dp[i][j] = 1e18;
            for (int k = opt[i][j - 1]; k <= min(j - 1, opt[i + 1][
                j]); k++) {
                long long cur = dp[i][k] + dp[k + 1][j] + cost(i, j
            );
            if (cur < dp[i][j]) { dp[i][j] = cur; opt[i][j] = k
            ; }
        }
    }
}

```

7.5 Sum Over Subsets DP (SOS DP)

Calculates $F(mask) = \sum_{sub \subseteq mask} A(sub)$ for all masks in $O(N \cdot 2^N)$ time complexity.

```

void solve_sos(int n, vector<int>& a) {
    vector<int> dp = a; // dp[mask] initially stores A[mask]
    for (int i = 0; i < n; ++i) {
        for (int mask = 0; mask < (1 << n); ++mask) {
            if (mask & (1 << i)) {
                dp[mask] += dp[mask ^ (1 << i)];
            }
        }
    }
}

```

To compute supersets instead ($F(mask) = \sum_{mask \subseteq super} A(super)$), change the condition to

check if the bit is NOT set: `if (!(mask & (1 << i)))` and transition via `dp[mask] += dp[mask ^ (1 << i)]`.

8 Geometry

8.1 Geometry Templates (Integer & Double)

Complete 2D geometry templates. Integer version for 100

```

struct Point {
    long long x, y;
    Point(long long x = 0, long long y = 0) : x(x), y(y) {}
    Point operator+(const Point& o) const { return {x + o.x, y + o.
        y}; }
    Point operator-(const Point& o) const { return {x - o.x, y - o.
        y}; }
};

long long dot(Point a, Point b) { return a.x * b.x + a.y * b.y; }
long long cross(Point a, Point b) { return a.x * b.y - a.y * b.x; }
long long dist2(Point a, Point b) { return dot(a - b, a - b); }

int ccw(Point a, Point b, Point a_c) {
    long long cr = cross(b - a, a_c - a);
    return cr == 0 ? 0 : (cr > 0 ? 1 : -1);
}

bool on_segment(Point p, Point a, Point b) {
    return ccw(a, b, p) == 0 && dot(a - p, b - p) <= 0;
}

bool intersect(Point a, Point b, Point a_c, Point d) {
    if (on_segment(a_c, a, b) || on_segment(d, a, b) || on_segment(
        a, a_c, d) || on_segment(b, a_c, d)) return true;
    return ccw(a, b, a_c) * ccw(a, b, d) < 0 && ccw(a_c, d, a) *
        ccw(a_c, d, b) < 0;
}

```

Listing 1: 1. INTEGER GEOMETRY TEMPLATE

```

const double EPS = 1e-9;

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    Point operator+(const Point& o) const { return {x + o.x, y + o.
        y}; }
    Point operator-(const Point& o) const { return {x - o.x, y - o.
        y}; }
    Point operator*(double c) const { return {x * c, y * c}; }
    Point operator/(double c) const { return {x / c, y / c}; }
};

double dot(Point a, Point b) { return a.x * b.x + a.y * b.y; }
double cross(Point a, Point b) { return a.x * b.y - a.y * b.x; }
double dist2(Point a, Point b) { return dot(a - b, a - b); }
double dist(Point a, Point b) { return sqrt(dist2(a, b)); }

int ccw(Point a, Point b, Point a_c) {
    double cr = cross(b - a, a_c - a);
    return abs(cr) < EPS ? 0 : (cr > 0 ? 1 : -1);
}

bool on_segment(Point p, Point a, Point b) {
    return ccw(a, b, p) == 0 && dot(a - p, b - p) <= EPS;
}

bool intersect(Point a, Point b, Point a_c, Point d) {
    if (on_segment(a_c, a, b) || on_segment(d, a, b) || on_segment(
        a, a_c, d) || on_segment(b, a_c, d)) return true;
    return ccw(a, b, a_c) * ccw(a, b, d) < 0 && ccw(a_c, d, a) *
        ccw(a_c, d, b) < 0;
}

```

```

}

Point line_intersection(Point a, Point b, Point a_c, Point d) {
    double cp = cross(b - a, d - a_c); assert(abs(cp) > EPS);
    return a + (b - a) * (cross(a_c - a, d - a_c) / cp);
}

```

Listing 2: 2. DOUBLE GEOMETRY TEMPLATE

8.2 Convex Hull (Andrew's Monotone Chain)

Computes 2D Convex Hull in $O(N \log N)$. Returns points in CCW order. Change `<= 0` to `< 0` to include collinear points.

```

long long cross(Point a, Point b, Point c) {
    return (b.x - a.x) * (c.y - a.y) - (b.y - a.y) * (c.x - a.x);
}

vector<Point> convex_hull(vector<Point>& pts) {
    int n = pts.size(), k = 0; if (n <= 3) return pts;
    vector<Point> hull(2 * n); sort(pts.begin(), pts.end());

    for (int i = 0; i < n; ++i) {
        while (k >= 2 && cross(hull[k - 2], hull[k - 1], pts[i]) <=
            0) k--;
        hull[k++] = pts[i];
    }
    for (int i = n - 2, t = k + 1; i >= 0; i--) {
        while (k >= t && cross(hull[k - 2], hull[k - 1], pts[i]) <=
            0) k--;
        hull[k++] = pts[i];
    }
    hull.resize(k - 1); return hull;
}

```

8.3 Incenter & Circumcenter of a Triangle

Computes the incenter and circumcenter of a triangle. Ensure points are not collinear.

```

Point get_incenter(Point A, Point B, Point C) {
    double a = dist(B, C), b = dist(A, C), c = dist(A, B), p = a +
        b + c;
    assert(p > 1e-9); return (A * a + B * b + C * c) / p;
}

Point get_circumcenter(Point A, Point B, Point C) {
    double d = 2 * (A.x * (B.y - C.y) + B.x * (C.y - A.y) + C.x * (
        A.y - B.y)); assert(abs(d) > 1e-9);
    double ax = A.x * A.x + A.y * A.y, bx = B.x * B.x + B.y * B.y,
        cx = C.x * C.x + C.y * C.y;
    double ux = (ax * (B.y - C.y) + bx * (C.y - A.y) + cx * (A.y -
        B.y)) / d;
    double uy = (ax * (C.x - B.x) + bx * (A.x - C.x) + cx * (B.x -
        A.x)) / d;
    return {ux, uy};
}

```

8.4 Li Chao Tree

Maximum query of lines on $[L, R]$ in $O(\log(R - L))$. For Minimum: init with `{0, INF}`, change comparison to `<`, and use `min()` in query.

```

struct Line {
    long long k, m;
    long long operator()(long long x) { return k * x + m; }
};

```

```

struct LiChaoTree {
    int n; vector<Line> tree;
    LiChaoTree(int n) : n(n), tree(4 * n, {0, -INF}) {}

    void insert(int node, int l, int r, Line newline) {
        int mid = (l + r) >> 1;
        bool bit_l = newline(l) > tree[node](l), bit_m = newline(
            mid) > tree[node](mid);
        if (bit_m) swap(tree[node], newline);
        if (l == r) return;
        if (bit_l != bit_m) insert(node << 1, l, mid, newline);
        else insert(node << 1 | 1, mid + 1, r, newline);
    }

    long long query(int node, int l, int r, int x) {
        if (l == r) return tree[node](x);
        int mid = (l + r) >> 1; long long res = tree[node](x);
        if (x <= mid) res = max(res, query(node << 1, l, mid, x));
        else res = max(res, query(node << 1 | 1, mid + 1, r, x));
        return res;
    }
};

```

8.5 LineContainer (CHT)

Dynamic Convex Hull Trick for maximum query in $O(\log N)$.
For Minimum: insert $\text{add}(-k, -m)$ and negate output $-\text{query}(x)$.

```

struct Line {
    mutable long long k, m, p;
    bool operator<(const Line& o) const { return k < o.k; }
    bool operator<(long long x) const { return p < x; }
};

struct LineContainer : set<Line, less<>> {
    static const long long INF = LLONG_MAX;

    long long div(long long a, long long b) { return a / b - ((a ^
        b) < 0 && a % b); }

    bool isect(iterator x, iterator y) {
        if (y == end()) { x->p = INF; return false; }
        if (x->k == y->k) x->p = x->m > y->m ? INF : -INF;
        else x->p = div(y->m - x->m, x->k - y->k);
        return x->p >= y->p;
    }

    void add(long long k, long long m) {
        auto z = insert({k, m, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->p >= y->p) isect(x,
            erase(y));
    }

    long long query(long long x) {
        assert(!empty()); auto l = *lower_bound(x); return l.k * x
            + l.m;
    }
};

```

8.6 Polygon Area (Shoelace Formula)

Computes the area of a polygon in $O(N)$ given its vertices ordered sequentially. Uses the global `Point` and `cross` from Geometry Template.

```

double polygon_area(const vector<Point>& pts) {
    long long area2 = 0; int n = pts.size();

```

```

    for (int i = 0; i < n; i++) {
        area2 += cross(pts[i], pts[(i + 1) % n]);
    }
    return abs(area2) / 2.0;
}

```

8.7 Smallest Enclosing Circle (Welzl's Algorithm)

Finds the minimum bounding circle in $O(N)$ expected time.
Uses global `Point`, `dist`, and `rng`.

```

struct Circle { Point center; double r; };

bool is_inside(Point p, Circle c) { return dist(p, c.center) <= c.r
    + 1e-9; }

Circle circle_from_3_points(Point A, Point B, Point C) {
    double bx = B.x - A.x, by = B.y - A.y, cx = C.x - A.x, cy = C.y
        - A.y;
    double BB = bx * bx + by * by, CC = cx * cx + cy * cy, D = 2 *
        (bx * cy - by * cx);
    Point center = {A.x + (cy * BB - by * CC) / D, A.y + (bx * CC -
        cx * BB) / D};
    return {center, dist(center, A)};
}

Circle circle_from_2_points(Point A, Point B) {
    return {{(A.x + B.x) / 2.0, (A.y + B.y) / 2.0}, dist(A, B) /
        2.0};
}

Circle welzl(vector<Point>& pts, vector<Point> r, int n) {
    if (n == 0 || r.size() == 3) {
        if (r.empty()) return {{0, 0}, 0};
        if (r.size() == 1) return {r[0], 0};
        if (r.size() == 2) return circle_from_2_points(r[0], r[1]);
        return circle_from_3_points(r[0], r[1], r[2]);
    }
    Point p = pts[n - 1]; Circle c = welzl(pts, r, n - 1);
    if (is_inside(p, c)) return c;
    r.push_back(p); return welzl(pts, r, n - 1);
}

Circle get_smallest_enclosing_circle(vector<Point> pts) {
    shuffle(pts.begin(), pts.end(), rng);
    return welzl(pts, {}, pts.size());
}

```

9 Misc

9.1 Custom Hash for hashmap & hashtable

```

struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }

    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM = chrono::steady_clock::
            now().time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};

```

To use this structure, pass `custom_hash` as the third template argument when defining your hash container:

```

unordered_map<long long, int, custom_hash> safe_map;
unordered_set<long long, custom_hash> safe_set;

```

9.2 Hill Climbing (Heuristics)

A local search meta-heuristic that continuously moves in the direction of increasing/decreasing value (ascending the hill) to find an optimal state. Unlike Simulated Annealing, it is purely greedy and can get trapped in local optima.

Core Methodology:

1. **Generate Initial State:** Construct a valid random configuration and calculate its objective cost function score.
2. **Evaluate Neighbors:** Generate small mutations or modifications to the active parameters (e.g., swapping two elements).
3. **Greedy Step:** Compare the mutated score with the original. If the mutant state improves the objective criteria, instantly accept it as the new baseline. Otherwise, discard it immediately.
4. **Termination:** Repeat until no neighboring modifications yield any further improvements (local peak reached), or the allocated execution time limit expires.

Strategies to Escape Local Optima (Random Restart): Because pure Hill Climbing stops at the very first local peak it encounters, combine it with a **Random Restart** strategy inside competitive programming constraints:

- Enclose the entire hill climbing block within an external loop or time-guard window (`while (clock() < 0.95 * CLOCKS_PER_SEC)`).
- Whenever a state becomes completely stagnant (no mutations improve the score anymore), save its peak metrics to a global maximum tracker.
- Fully randomize the current state configuration back to zero and launch a brand new hill climbing ascent from that arbitrary coordinate.

9.3 Simulated Annealing (Heuristics)

Randomized optimization for NP-hard problems. Escapes local minima by accepting worse states with a temperature-dependent probability $P = e^{-\Delta/T}$.

```

inline double get_rand() { return (double)rng() / mt19937::max(); }

struct State {
    long long cost;
    void init() {} // Generate initial state & compute cost
    void mutate() {} // Apply small random modification & update cost
    void revert() {} // Undo mutation if rejected
};

void simulated_annealing() {
    State curr; curr.init();
    State best = curr;

    double T = 1e5, T_min = 1e-3, alpha = 0.9995; // Adjust parameters based on problem
    auto start_time = chrono::steady_clock::now();

    while (T > T_min) {
        // TLE Protection: Exit just before 1.0s time limit
        auto elapsed = chrono::duration<double>(chrono::steady_clock::now() - start_time).count();
        if (elapsed > 0.95) break;

        long long old_cost = curr.cost;
        curr.mutate();
        long long diff = curr.cost - old_cost; // Negative means improvement (Minimization)

        // Metropolis Criterion: Accept if better, or with probability exp(-diff / T)
        if (diff < 0 || exp(-diff / T) > get_rand()) {
            if (curr.cost < best.cost) best = curr;
        } else {
            curr.revert();
        }
        T *= alpha; // Cool down
    }
}

```

9.4 ModInt Template

```

template<int MOD>
struct ModInt {
    int v;
    ModInt() : v(0) {}
    ModInt(long long _v) { v = int((-MOD < _v && _v < MOD) ? _v : _v % MOD); if (v < 0) v += MOD; }

    explicit operator int() const { return v; }
    bool operator==(const ModInt& o) const { return v == o.v; }
    bool operator!=(const ModInt& o) const { return v != o.v; }

    ModInt& operator+=(const ModInt& o) { v += o.v; if (v >= MOD) v -= MOD; return *this; }
    ModInt& operator-=(const ModInt& o) { v -= o.v; if (v < 0) v += MOD; return *this; }
    ModInt& operator*=(const ModInt& o) { v = int((long long)v * o.v % MOD); return *this; }

    ModInt pow(long long p) const {
        ModInt res = 1, base = *this;
        for (; p > 0; p >>= 1, base *= base) if (p & 1) res *= base;
        return res;
    }
    ModInt inv() const { return pow(MOD - 2); } // Requires MOD to be a prime
    ModInt& operator/=(const ModInt& o) { return *this *= o.inv(); }
}

```

```

friend ModInt operator+(ModInt a, const ModInt& b) { return a += b; }
friend ModInt operator-(ModInt a, const ModInt& b) { return a -= b; }
friend ModInt operator*(ModInt a, const ModInt& b) { return a *= b; }
friend ModInt operator/(ModInt a, const ModInt& b) { return a /= b; }
};
using mint = ModInt<1000000007>;

```

- Use mint to declare variables. For example: mint a = 5, b = 10;
- Supports all standard arithmetic operators (+, -, *, /), compound assignments (+=, -=, *=, /=), and comparisons (==, !=).
- Call a.pow(p) to compute $a^p \pmod{MOD}$ efficiently in $O(\log p)$.
- Call a.inv() to find the modular multiplicative inverse using Fermat's Little Theorem.

9.5 Policy-Based Data Structures (PBDS)

An extension structure in GCC that provides a high-performance Ordered Set, supporting index-based lookups (find_by_order) and order queries (order_of_key) in $O(\log N)$.

```

#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;

// Definition for Ordered Set
template <typename T>
using ordered_set = tree<T, null_type, less<T>, rb_tree_tag, tree_order_statistics_node_update>;

// Definition for Ordered Multiset (allows duplicate keys)
template <typename T>
using ordered_multiset = tree<T, null_type, less_equal<T>, rb_tree_tag, tree_order_statistics_node_update>;

```

Basic operations and functions usage pattern:

```

ordered_set<int> st;

// 1. Insert elements normally
st.insert(1); st.insert(4); st.insert(8);

// 2. find_by_order(k): Returns an iterator to the k-th smallest element (0-indexed)
auto it = st.find_by_order(1); // Points to '4'
int val = *it;

// 3. order_of_key(x): Returns the number of elements strictly smaller than x
int cnt = st.order_of_key(5); // Returns 2 (elements 1 and 4 are smaller)

// 4. Erase element in ordered_multiset (MUST use iterator to avoid removing all duplicates)
ordered_multiset<int> mst;
mst.insert(4); mst.insert(4);
mst.erase(mst.find_by_order(mst.order_of_key(4))); // Safely erases only one '4'

```

9.6 Stress Tester

Automated stress testing harness using identical execution checking paths. Run main.cpp to compile and test.

```

int32_t main() {
    ifstream f_user("user.out"), f_brute("brute.out");
    string s_user, s_brute;
    while (f_user >> s_user) {
        if (!(f_brute >> s_brute) || s_user != s_brute) return 1;
    }
    if (f_brute >> s_brute) return 1;
    return 0;
}

```

Listing 3: checker.cpp

```

mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());

int rand_range(int l, int r) {
    uniform_int_distribution<long long> dist(l, r);
    return dist(rng);
}

void gen_test() {
    ofstream out("test.in");
    out << rand_range(1, 100) << " " << rand_range(1, 100) << "\n";
}

int32_t main() {
    system("g++ -O3 user.cpp -o user");
    system("g++ -O3 brute.cpp -o brute");
    system("g++ -O3 checker.cpp -o checker");

    for (int t = 1; t <= 10000; ++t) {
        gen_test();
        system("./user < test.in > user.out");
        system("./brute < test.in > brute.out");
        if (system("./checker")) {
            cout << "WA on test " << t << "\n";
            return 0;
        }
    }
    cout << "All tests passed!\n";
    return 0;
}

```

Listing 4: main.cpp

10 Classic

10.1 Bounded Knapsack with Binary Grouping

Solves the Bounded Knapsack problem where each item i has a limited count C_i . Reduces the state space to a 0-1 Knapsack instance in $O(W \sum \log C_i)$ time complexity.

```

struct Item {
    long long w, v;
};

long long bounded_knapsack(int n, long long W, const vector<long long>& weight, const vector<long long>& value, const vector<int>& cnt) {
    vector<Item> items;
    for (int i = 0; i < n; ++i) {
        int rem = cnt[i];
        for (int k = 1; rem > 0; k <= 1) {
            int take = min(k, rem);

```

```

        rem -= take;
        items.push_back({take * weight[i], take * value[i]});
    }

    vector<long long> dp(W + 1, 0);
    for (const auto& item : items) {
        for (int c = W; c >= item.w; --c) {
            dp[c] = max(dp[c], dp[c - item.w] + item.v);
        }
    }
    return dp[W];
}

```

10.2 CSES Counting Tilings

Solves the grid tiling problem (CSES 2181) for an $N \times M$ grid using 1×2 and 2×1 dominoes in $O(N \cdot M \cdot 2^N)$ time and $O(2^N)$ space complexity.

```

const int MOD = 1e9 + 7;

int solve_tiling(int n, int m) {
    if (n > m) swap(n, m); // Optimize for the smaller dimension
    vector<long long> dp(1 << n, 0);
    dp[0] = 1;

    for (int j = 0; j < m; ++j) {
        for (int i = 0; i < n; ++i) {
            vector<long long> next_dp(1 << n, 0);
            for (int mask = 0; mask < (1 << n); ++mask) {
                if (!dp[mask]) continue;

                if (mask & (1 << i)) {
                    // Current cell is already covered by a
                    vertical domino from the top row
                    next_dp[mask ^ (1 << i)] = (next_dp[mask ^ (1
<< i)] + dp[mask]) % MOD;
                } else {
                    // Option 1: Place a horizontal domino (extends
                    to the right column)
                    next_dp[mask ^ (1 << i)] = (next_dp[mask ^ (1
<< i)] + dp[mask]) % MOD;

                    // Option 2: Place a vertical domino down (if
                    not on the last row and bottom cell is empty)
                    if (i + 1 < n && !(mask & (1 << (i + 1)))) {
                        next_dp[mask ^ (1 << (i + 1))] = (next_dp[
mask ^ (1 << (i + 1))] + dp[mask]) % MOD;
                    }
                }
            }
            dp = move(next_dp);
        }
    }
    return dp[0];
}

```

10.3 Closest Pair of Points

Finds the closest pair of points in $O(N \log N)$ time using a sweep-line algorithm with `std::set`. Uses global `Point` and `dist` from Geometry Template.

```

bool compare_y(const Point& a, const Point& b) { return a.y < b.y;
}

pair<Point, Point> closest_pair(vector<Point>& pts) {
    int n = pts.size(); sort(pts.begin(), pts.end());
}

```

```

double min_d = 1e18; pair<Point, Point> best_pair;
set<Point, decltype(&compare_y)> active_set(&compare_y);

int left = 0;
for (int i = 0; i < n; i++) {
    while (pts[i].x - pts[left].x >= min_d) active_set.erase(
        pts[left++]);
    Point lower_bound_pt = {0, (long long)(pts[i].y - min_d)};
    auto it = active_set.lower_bound(lower_bound_pt);

    while (it != active_set.end() && it->y - pts[i].y < min_d) {
        double d = dist(pts[i], *it);
        if (d < min_d) { min_d = d; best_pair = {pts[i], *it};
        }

        it++;
    }
    active_set.insert(pts[i]);
}
return best_pair;
}

```

10.4 De Bruijn Sequence

Generates the shortest cyclic sequence of length B^K that contains every possible sequence of length K over an alphabet of size B as a substring exactly once.

```

struct DeBruijn {
    int b, k;
    vector<int> sequence, a;

    DeBruijn(int _b, int _k) : b(_b), k(_k), a(_k * _b, 0) {}

    void db_dfs(int t, int p) {
        if (t > k) {
            if (k % p == 0) {
                for (int i = 1; i <= p; ++i) sequence.push_back(a[i
]);
            }
            else {
                a[t] = a[t - p];
                db_dfs(t + 1, p);
                for (int i = a[t - p] + 1; i < b; ++i) {
                    a[t] = i;
                    db_dfs(t + 1, t);
                }
            }
        }

        vector<int> generate() {
            sequence.clear();
            db_dfs(1, 1);
            return sequence;
        }
    };
}

```

11 Formulas

11.1 Essential Theorems

Critical theorems and properties for structural reductions and mathematical transformations.

Graph Theory Invariants & Reductions:

- **Dilworth:** Max antichain (no two comparable) = min chains to cover poset. Reduces to Max Bipartite Matching.

- **Konig:** In any bipartite graph, $|Max\ Matching| = |Min\ Vertex\ Cover|$.
- **Independent Set:** S is vertex cover iff $V \setminus S$ is independent set
 $\implies |Min\ Vertex\ Cover| + |Max\ Independent\ Set| = |V|$.
- **Gallai:** In graphs without isolated nodes, $|Max\ Matching| + |Min\ Edge\ Cover| = |V|$.
- **Eulerian Path:** Undirected: cycle iff all degrees even; path iff exactly 0 or 2 nodes have odd degree. Directed: cycle iff $in[u] = out[u]$ for all u ; path iff at most one node has $out - in = 1$ and at most one has $in - out = 1$.

Number Theory Identities:

- **Euler & Fermat:** $\gcd(a, m) = 1 \implies a^{\phi(m)} \equiv 1 \pmod{m}$. If p is prime, $a^{p-1} \equiv 1 \pmod{p}$.
- **Bezout:** $\exists x, y$ such that $ax + by = \gcd(a, b)$, solved via ExtGCD.
- **Wilson:** Integer $n > 1$ is prime iff $(n-1)! \equiv -1 \pmod{n}$.
- **Mobius Inversion:** If $g(n) = \sum_{d|n} f(d)$, then $f(n) = \sum_{d|n} \mu(d)g(n/d)$.
- **GCD Properties:**
 - $\gcd(a^n - 1, a^m - 1) = a^{\gcd(n, m)} - 1$.
 - $\gcd(F_a, F_b) = F_{\gcd(a, b)}$

11.2 Essential Formulas

Combinatorics & Modular Queries: $C_n^k = \frac{n!}{k!(n-k)!} \pmod{M}$. Precompute factorials in $O(N)$. Linear inverse:

$inv[i] = -(M / i) * inv[M \% i] \% M + M$;

Vandermonde's Identity: $\sum_{k=0}^r C_m^k C_n^{r-k} = C_{m+n}^r$ (Choosing r items from two combined groups).

Fermat's Polygonal Number (Gauss EUREKA): Every $N > 0$ can be expressed as the sum of at most k polygonal numbers of side k (max count = k). General formula for n -th polygonal number of side k :

$$P_k(n) = \frac{(k-2)n^2 - (k-4)n}{2} = n + (k-2) \cdot \frac{n(n-1)}{2}$$

- **Recurrence Relation between Sides:** A k -polygonal number can be derived from a $(k-1)$ -polygonal number by adding the $(n-1)$ -th triangular number ($T_{n-1} = \frac{n(n-1)}{2}$):

$$P_k(n) = P_{k-1}(n) + T_{n-1} = P_{k-1}(n) + \frac{n(n-1)}{2}$$

- **Triangular Decomposition:** Any k -polygonal number can be broken down directly into a combination of the n -th and $(n - 1)$ -th triangular numbers:

$$P_k(n) = T_n + (k - 3) \cdot T_{n-1}$$

- **General $O(1)$ Check / Inverse Function:** To check if N is a k -polygonal number, solve $(k - 2)n^2 - (k - 4)n - 2N = 0$. Let $\Delta = (k - 4)^2 + 8N(k - 2)$. N is valid iff Δ is a perfect square and $n = \frac{(k-4)+\sqrt{\Delta}}{2(k-2)}$ is an integer.
- **Find side k from N and n :** $k = 2 + \frac{2(N-n)}{n(n-1)}$.

Catalan Numbers: (BSTs, valid parenthesization, non-crossing polygon dissections).
 $C_0 = 1, C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i} = \frac{1}{n+1} C_{2n}^n = C_{2n}^n - C_{2n}^{n-1}$ Terms: 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862.

Derangements: (Permutations with no fixed points).
 $D_1 = 0, D_2 = 1, D_n = (n - 1)(D_{n-1} + D_{n-2}) = nD_{n-1} + (-1)^n$
Terms: 0, 1, 2, 9, 44, 265, 1854, 14833, 133496.

Stirling Numbers of the 2nd Kind: (Partition n labeled items into k unlabeled non-empty subsets).
 $S(n, k) = S(n - 1, k - 1) + k \cdot S(n - 1, k)$. Base:
 $S(0, 0) = 1, S(n, 0) = S(0, n) = 0$. Partition n labeled items into k labeled boxes = $k! \cdot S(n, k)$.

Lucas’ Theorem: $C_n^k \pmod p$ for small prime p ($n, k \leq 10^{18}$).
Base- p representations:
 $n = \sum n_i p^i, k = \sum k_i p^i \implies C_n^k \equiv \prod C_{n_i}^{k_i} \pmod p$ (where $C_{n_i}^{k_i} = 0$ if $n_i < k_i$).

Burnside’s Lemma: Number of orbits (distinct objects under symmetry) = $\frac{1}{|G|} \sum_{g \in G} |X^g|$ where $|G|$ is size of symmetry group, and $|X^g|$ is number of elements fixed by symmetry g .

Graph Theory Counting:

- **Cayley’s Formula:** Labeled spanning trees in $K_n = n^{n-2}$. In a forest of k trees with roots $r_1..r_k$ covering all n nodes, ways to connect them into a single tree = $n^{k-2} \cdot \prod sz(T_i)$.
- **Euler’s Planar Formula:** Connected planar graph: $V - E + F = 2$.
- **Matrix Tree Theorem:** Let A be adjacency matrix, D degree matrix. Kirchhoff matrix $L = D - A$. Delete row i and col i . Determinant of resulting matrix = number of spanning trees.

Geometry Theorems:

- **Pick’s Theorem:** Grid polygon area $A = I + B/2 - 1$. (I : interior points, B : boundary points).
- **Boundary Points:** Integer points on segment (x_1, y_1) to (x_2, y_2) inclusive = $\gcd(|x_1 - x_2|, |y_1 - y_2|) + 1$.

11.3 Cheatsheet

A comprehensive reference table for prime counts, highly composite numbers, primorials, factorials, and modular bounds.

1. Number of Primes $\pi(n)$:

n	10^2	10^3	10^4	10^5	10^6	10^7	10^8	10^9	10^{10}
$\pi(n)$	25	168	1229	9592	78498	664579	5761455	50847534	455052511

2. Highly Composite Numbers (Max divisors $d(x)$ within range):

$\leq n$	10^3	10^4	10^5	10^6	10^7	10^8	10^9
x	840	7560	83160	720720	8648640	73513440	735134400
$d(x)$	32	64	128	240	448	768	1344

$\leq n$	10^{10}	10^{11}	10^{12}	10^{13}	10^{14}	10^{15}	10^{16}	10^{17}	10^{18}
$d(x)$	2304	4032	6720	10752	17280	26880	41472	64512	103680

3. Primorials ($n\#$ - Product of primes $\leq n$):

n	2	3	5	7	11	13	17	19	23	29
$n\#$	2	6	30	210	2310	30030	510510	9699690	223092870	6469693230

4. Factorials ($n!$):

$4!$	$5!$	$6!$	$7!$	$8!$	$9!$	$10!$	$11!$	$12!$	$13!$
24	120	720	5040	40320	362880	3628800	39916800	479001600	6227020800

5. Hashing Bases & NTT Friendly Modulos:

Type	Value (Prime)	Primitive Root (g)	Max NTT Size
String Hash Base	31, 37, 131, 13331	—	—
Standard Mod	$10^9 + 7$ (1000000007)	5	Non-NTT
Standard Mod	$10^9 + 9$ (1000000009)	13	Non-NTT
NTT Modulo	998244353	3	2^{23}
NTT Modulo	167772161	3	2^{25}
NTT Modulo	469762049	3	2^{26}

6. Data Type Limits & Overflow Thresholds:

Type	Exact Max Value	Approx.	Safe Geometry Coordinate
32-bit int	2, 147, 483, 647	$\approx 2 \times 10^9$	$X, Y \leq 30\,000$
64-bit long long	9, 223, 372, 036, 854, 775, 807	$\approx 9 \times 10^{18}$	$X, Y \leq 2 \times 10^9$
<code>__int128</code>	170, 141, 183, 460, 469, 231, 731, ...	$\approx 1.7 \times 10^{38}$	Max cross product

7. Fibonacci & Catalan Number Limits:

Sequence	Largest Index fitting 32-bit int	Largest Index fitting 64-bit long long
Fibonacci (F_n)	$F_{46} = 1,836,311,903$	$F_{92} = 7,540,113,804,746,346,429$
Catalan (C_n)	$C_{19} = 1,767,263,190$	$C_{35} = 31,162,854,949,073,012,624$

- Highly Composite Numbers tables are crucial for finding numbers with the maximum number of divisors under a specific upper bound.
- Max values for standard types: 13! fits within standard 32-bit signed [int](#), while 20! (2 432 902 008 176 640 000) is the largest factorial fitting inside a 64-bit signed [long long](#).